

SEPARATING CHOICE FROM VALIDITY: A VALIDATED SEARCH-CHOICE MEASUREMENT FOR VISION-LANGUAGE SEARCH POLICIES

Anonymous authors

Paper under double-blind review

ABSTRACT

We ask whether search choice can be measured separately from operation validity for vision-language search policies. On additive best-first cells of our enumeration contract, where expansions submit complete candidate sets, random-valid is decision-identical to the exact reference on 48/48 pairs, so success there measures validity, not choice. With submission-order heap serials on the same runtime, 19/48 pairs diverge, so the check can return either verdict, but the pre-registered majority prediction failed. On a redesigned choice-frontier contract and fresh 12-task validation panel, the pre-registered ladder test passes and replicates on a held-out panel. The solve-versus-budget area M1 orders the exact reference (0.875) above exact- ϵ selectors choosing uniformly on 25%, 50%, and 75% of decisions (0.793, 0.603, 0.347) and random-valid (0.021), with every adjacent 95% lower bound positive (smallest gap +0.082 [+0.051, +0.111]). The rungs read the privileged heap head, so the ladder validates the ordering of the measurement, not any realisable policy.

A scaled adapter beats uniform choice on D_3 , adapter minus random-valid M1 averaged over three seeds (+0.285 [+0.140, +0.433]), and on a held-out panel (+0.416 [+0.250, +0.591]). On an unscreened panel whose tasks random-valid never solved in screening, the gain is +0.149 [+0.030, +0.283] and rests on 4 of 12 tasks (descriptive). It is interval-separated above the exact- ϵ 0.75 rung on the held-out panel (+0.227 [+0.053, +0.403]) but not separated from that rung on the validation panel (−0.041 [−0.187, +0.102]), the unscreened panel, or pooled over 35 tasks (+0.058 [−0.040, +0.156], descriptive). On ScienceWorld random-valid registers choice (success 0.915 versus 0). On LLM-First Search it diverges from the reference in every pair yet solves every task under the paper’s token budget (1.0 versus 0.825 for a substituted reference model), and a decision-matched budget restores the contrast (0.0775). The complete-set identity failure is shown only on our runtime. The choice-quality claim is development-stage, holds relative to uniform choice and the exact- ϵ ladder on three panels with three training seeds, and carries no planning-ability or generality claim.

1 INTRODUCTION

On classical planning probes (Valmeekam et al., 2023b; Stein et al., 2025; Kokel et al., 2026), off-the-shelf language models improve at operational reasoning with scale and prompting but not at structural enumeration (Huang et al., 2026), leaving open whether a model can be trained to sustain a declared search procedure. We train a Search Process Policy (Section 3) to emit Typed Search Operations that a no-repair Trusted Search Runtime validates against Algorithm Invariants, over BFS, width-based, and additive best-first search under text, visual, and multimodal observations (Section 4). Success on such an interface can reflect operation validity or choice among valid operations. A random-valid control, which picks uniformly among valid operations, separates the two only when its decisions can differ from the reference’s, so we ask whether search choice can be measured separately from operation validity.

On the additive best-first cells of our enumeration contract, where an expansion submits its complete candidate set and heap serials follow the sorted candidate order, random-valid is decision-identical to

the exact reference on 48/48 pairs, so success there measures operation validity rather than choice. On the same runtime with heap serials assigned in submission order, the same check finds 19/48 divergent pairs, so the check can return either verdict. The pre-registered prediction of divergence on most pairs (more than 24/48) did not hold (Section 5.2). The redesigned choice-frontier contract (Figure 1) makes the policy select which runtime-owned frontier state to expand under a binding budget (Section 5.3). On a fresh 12-task panel from 7 domains, the pre-registered ladder test passes. The solve-versus-budget area M1 orders the exact reference (0.875) above exact- ϵ selectors that choose uniformly on 25% (0.793), 50% (0.603), and 75% (0.347) of decisions, and random-valid (0.021) lowest, and every adjacent paired task-clustered 95% lower bound is positive (smallest gap +0.082 [+0.051, +0.111]). The rungs read the privileged heap head, so the ladder validates the ordering of the measurement, not any realisable policy.

Across three training seeds the adapter beats uniform frontier choice on the validation panel, $D_3 = +0.285$ [+0.140, +0.433], where D_3 is adapter minus random-valid M1 averaged over three seeds, with every seed positive (+0.328, +0.229, +0.297), which meets the pre-registered POSITIVE rule, and the result replicates on the held-out panel P2, $D_3 = +0.416$ [+0.250, +0.591]. It is interval-separated above the exact- ϵ 0.75 rung on the held-out panel (+0.227 [+0.053, +0.403]) but not separated from that rung on the validation panel (−0.041 [−0.187, +0.102]), the unscreened panel, or pooled over 35 tasks (+0.058 [−0.040, +0.156], descriptive, Section 5.4). On an unscreened panel whose tasks random-valid never solved in screening, the gain is +0.149 [+0.030, +0.283] and rests on 4 of 12 tasks (descriptive). The gains on all three panels are obtained by imitating the exact reference. On ScienceWorld (Wang et al., 2022), a published sequential-choice interface, random-valid diverges from the gold trajectory at the first decision in 1000/1000 pairs, with success 0.915 versus 0 for random-valid over 200 tasks, so success there measures choice. The audit covers 6 of its 30 task types, and none of the seven published interfaces we surveyed uses a complete-set submission contract, so the complete-set identity failure is shown only on our runtime. On LLM-First Search (Herr et al., 2025), random-valid diverges in all 400 pairs yet solves all 80 tasks under the paper’s token budget (1.0 versus 0.825 for a substituted reference model), and only 0.0775 at the reference’s decision count, so success saturates unless the budget binds decisions (Section 5.5). Under the enumeration contract, process SFT solved 125 of 288 baseline entries, all in additive cells, and 0/24 in each BFS and BFWS cell (Section 5.6).

This paper makes four contributions.

- **A per-decision identity audit that returns both verdicts on one runtime.**
- **A choice-frontier measurement validated by a pre-registered exact- ϵ ladder.**
- **A learned result above uniform choice on three panels, with its boundaries.**
- **An external audit of two published interfaces, one where random-valid registers choice and one where its success saturates under the paper’s own budget.**

All enumeration-contract cells are single-seed except two replication cells (seeds 17/29/71, descriptive $n=3$). The choice-quality claim is development-stage, holds relative to uniform choice and the exact- ϵ ladder on three panels with three training seeds, and carries no planning-ability or generality claim. Held-out here means the fresh panel P2, frozen before any evaluation episode from seeds on which no adapter had been evaluated (the seed extension followed a controls-only screen of the first 40 seeds), and the frozen 45-task held-out final evaluation remains unexecuted. The full-scope v1 generalization and second-backbone branches were withdrawn after their terminal VALID_STOP receipts (closed 2026-09-21).

2 RELATED WORK

Planning benchmarks (Valmeekam et al., 2023a;b; Stein et al., 2025; Kokel et al., 2026) score plans or plan judgments rather than step-by-step execution of a declared algorithm such as A* (Hart et al., 1968) or BFWS (Lipovetzky & Geffner, 2017), and trace-trained models (Lehnert et al., 2024; Gandhi et al., 2024) and step imitators (Veličković et al., 2020; Zaremba & Sutskever, 2014) lack typed operations and in-episode validation. The choice-frontier contract is a learned node-selection policy in the sense of policy-guided heuristic search (Orseau & Lelis, 2021), and the submission-order counterfactual isolates a heap tie-breaking effect of the kind studied for A* search (Asai & Fukunaga, 2016).

In Tree of Thoughts (Yao et al., 2023) and RAP (Hao et al., 2023) a scaffold or UCT policy owns exploration, whereas our model issues the operations that determine it, and Scratchpads (Nye et al., 2021), per-step labels (Lightman et al., 2024), Visual Sketchpad (Hu et al., 2024), and MVoT (Li et al., 2025) supervise or generate unchecked free-form steps and imagery rather than attributing control to an observation channel. Unlike repair loops (Kambhampati et al., 2024; Madaan et al., 2023) and syntax-only constrained decoding (Scholak et al., 2021; Willard & Louf, 2023), our runtime never repairs and checks algorithmic invariants on every operation.

Evaluation-validity work in planning and reasoning warns that common-sense tasks can confound planning with retrieval (Valmeekam et al., 2023a), and embodied suites such as BEHAVIOR-1K broaden task coverage (Li et al., 2024). Research on multimodal training documents VQA models relying on language priors rather than image content (Zhu et al., 2020), and video-language evaluation reports selection bias toward answer positions (Loginova et al., 2025). Label-position selection bias is documented for LLM multiple-choice selectors (Zheng et al., 2024), and the fixed-versus-random-position distractor test is the corresponding unrun diagnostic for menu-based search. LVLM chain-of-thought audits find that visual or format cues can influence answers without appearing in reasoning traces (Balasubramanian et al., 2025). Process supervision supplies feedback on intermediate reasoning steps (Lightman et al., 2024), but that training signal alone does not test whether a grounded action menu makes frontier choices identifiable. Text-game work treats access to valid actions as an evaluation handicap. Jericho lists world-change detection for identifying valid actions among its optional handicaps, and CALM reports that on half of its unseen games it is competitive with or better than models with access to ground-truth admissible actions (Hausknecht et al., 2020; Yao et al., 2020). ScienceWorld (Wang et al., 2022) is a published sequential-choice interface that likewise exposes a per-step choice among valid actions, so a random-valid control there registers choice. Our identity audit checks this property per decision, and the complete-set submission contract where it fails was not found among the surveyed published interfaces. The menu manipulations in this paper probe this interface, and only the unrun fixed-versus-random-position variant would separate surface-following from state grounding. This paper operationalizes the known admissible-action handicap in search-execution evaluation with a per-decision identity audit that returns both verdicts on one runtime and anchors the validated choice measurement of Section 5.

3 SEARCH PROCESS POLICY

An episode fixes (x, A, m, B) : a task $x = (D, s_0, G)$ with deterministic grounded domain D , a declared algorithm A given as input, a modality adapter m , and a frozen budget B . At step t the policy sees $o_t = \phi_m(c_t)$, a text-state, visual-state, or multimodal-state rendering of the decision context that must be decision-sufficient (exposing every fact the exact teacher uses at t), with Search Memory and candidates kept textual.

Typed Search Operation. The policy emits $u_t = (\tau_t, \theta_t)$, a type from A ’s vocabulary plus every exploration-determining operand, and an emission that omits one is invalid. **Search Memory.** M_t is bounded runtime-held state (frontier, visited table, BFWS novelty counters), not model state, and o_t is the policy’s only channel to history. **Algorithm Invariant.** Deterministic per-algorithm properties are checked on every operation, namely FIFO order (BFS), novelty and goal-count priority (BFWS (Lipovetzky & Geffner, 2017)), and $g + 3h_{\text{add}}$ or h_{add} order under a declared tie-break (weighted A* (Hart et al., 1968; Pohl, 1970) and greedy best-first over h_{add} (Bonet & Geffner, 2001)), and compliance establishes neither termination, completeness, optimality, nor success. **Trusted Search Runtime.** The runtime only validates, applies, persists, or rejects, never choosing omitted operands, reordering candidates, or repairing, so raw emissions replay to the same trace (schemas in the Operation Schemas appendix, Appendix A, invariants in the Algorithm Invariant Definitions appendix).

Process supervision is token-level cross-entropy cloning of the teacher’s operation on Search-Trace Segments, bounded slices of replay-verified exact traces collected in a dataset $\mathcal{D}$, which optional DAGger (Ross et al., 2011) extends with teacher corrections on visited contexts.

4 EXPERIMENTAL DESIGN

Each algorithm-by-modality-by-condition cell asks whether a trained Search Process Policy exceeds an oracle-assisted validity control on full-episode budgeted success (Section 3), under frozen pro-

protocols, a 336 GPU-hour cap, and prespecified gate receipts. Panels are frozen and outcome-blind, the 45-task held-out manifest has never been accessed, and every learned cell uses one training seed (17) except two replicated cells (seeds 29 and 71 added) and the choice-frontier adapter, which has three training seeds (17, 29, 71).

Each cell holds the pretrained base, process SFT, and process SFT plus two DAgger iterations (Ross et al., 2011) beside an exposure-matched continued-SFT control. SFT trains a fresh LoRA adapter (Hu et al., 2022) on 512 replay-verified records for one epoch (16 updates), except the BFWS gate checkpoint, which trained on the full 47,780-record corpus for two epochs. The backbone is Qwen3-VL-8B-Instruct (Bai et al., 2025) (revision 0c351dd0), with InternVL3.5-8B (Wang et al., 2025) as second backbone.

Exact-classical runs the declared algorithm and defines the imitation target, random-valid proposes only runtime-valid operations, neither is a learned ability, and when a reference saturates a panel the learned-to-control gain there is unmeasurable rather than zero. Under the enumeration contract, random-valid reads the same runtime-built model input as the policy and, after filtering, draws one operation per decision uniformly at random with a seeded pseudo-random generator from the listed candidates, which are successor-candidate rows under the additive algorithms, non-duplicate evaluated successors with an admissible declared frontier position under BFWS, and unvisited successors under BFS, with a frontier-retire operation under BFWS and BFS when none remains. The sampled operation is submitted once through the same runtime path as a model output, with no retry and no repair. Under the choice-frontier contract the control is uniform menu choice of one offered frontier label per decision under a seeded generator, again with no retry or repair, so the two definitions share a name but not a distribution and each random-valid number is read under the contract of its row.

Splits are by whole task instance with audited corpora (Data and Split Procedures appendix). The modality matrix evaluates 24 tasks from 12 domains under three observations, giving 288 episodes per arm. Budgets count decision calls with invalid operations charged. Learned adapters are capped at twice the exact-reference decision count, the base is priced at one call, and an episode ends at its first invalid emission and counts only after independent replay reproduces it. Contrasts use a paired whole-instance bootstrap (10,000 resamples, seed 1729, 95% percentile) unless a protocol states otherwise (Statistics appendix), and transfer uses exact McNemar tests with Holm correction across 36 comparisons.

The two contracts are compared in Figure 1 and Table 25. The choice-frontier contract has as its pre-registered pass condition the per-pair identity audit of random-valid against exact reference over 18 seed-17 pairs (CHOICE_SENSITIVE on any divergence, ZERO_DECISION_HEADROOM otherwise, no re-runs), also run as a diagnostic on the enumeration contract’s additive cells.

Follow-up protocols (the Choice-Frontier Contract appendix and the Results Detail appendices) were each frozen before their own first outcome, on separate ledgers, but after earlier outcomes on the same nine tasks (Section 5.6). They add two image-only native arms that keep only the grounded, unscored menu as text (`visual-nomem-state`, and `visual-seq-state` with $K=8$ predecessor scenes) and menu manipulation by seeded permutation and injection of up to $k=3$ schema-valid inapplicable actions.

The first choice-frontier evaluation reuses the frozen 9-task native-arms membership (8 domains, C^* from 3 to 6) under both additive algorithms. The training corpus was derived on CPU by replaying the frozen exact traces through the new contract and verified expansion-for-expansion against them (58 episodes bit-exact).

The CPU-only exploratory comparator zoo tests Pareto dominance over M1 (solve-versus-budget area over multipliers 1 to 2), M2 (expansion overhead), and M3 (plan cost) against five comparators under a fifteen-test Holm family, with selectors that read privileged g/h scalars or PDDL atoms.

The ladder protocol, frozen before task generation on a fresh 12-task panel, passes only if all four adjacent paired task-cluster 95% lower bounds are positive across the exact reference, exact- ϵ rungs 0.25, 0.50, and 0.75, and random-valid (12-task bootstrap, seed 133, 10,000 draws). The scaled-adapter protocol, frozen before task generation, fixes the estimand D , learned minus random-valid M1, and a POSITIVE rule with materiality margin 0.05 and equivalence margin ± 0.05 . Two follow-up protocols, each frozen before its outcomes, replicate the adapter over three seeds and on two fresh panels, one screened and held out (P2) and one unscreened (P2u), and held-out refers to P2 only.

Two further protocols, a zero-shot second policy and a second external audit, were also frozen before their outcomes.

At submission (receipts in Table 2), the BFS pilot closed as history with VALID_STOP, the BFWS gate passed on its development panel, and the modality matrix, DAgger, failure calibration, corruption, transfer, seed replication, and all follow-ups executed, with generalization and second backbone at reduced v2 scope only. The BFWS-to-BFS panel overlap is not pinned in the retained manifests.

5 RESULTS

This section reports the executed evidence in gate order, with success, validity, invariant compliance, invalid-operation charges, budget, and learned-to-control gain kept as separate measurements and operational detail in the Results Detail appendices. The random-valid and exact-classical references are oracle-assisted bounds, not learned abilities, and where they saturate they bound what the comparison can measure rather than certify policy quality. Every panel is a development panel, every enumeration-contract learned arm rests on one training seed (17) except two replication cells (Section 5.6), the choice-frontier adapter has three training seeds, and the frozen 45-task held-out final evaluation has not been run, so no finding here is a final comparative efficacy claim.

5.1 GATE RECEIPTS AND EXECUTED SCOPE

The expanded study program spent 68.5244 of its 336 GPU-hour cap (2 A100 80GB GPUs), and every executed branch, follow-up window, and CPU-only analysis reconciles with complete coverage and independent replay, including 1,152/1,152 baseline episodes. Five of seven admitted branches completed at v1 scope, and two (second backbone, generalization/robustness) stopped with terminal VALID_STOP receipts at their v1 cost gates and completed only at reduced scope (Section 5.6). Before the program, the BFS pilot closed with a governed VALID_STOP because random-valid also reached 1.0, the BFWS structural gate passed, and the 45-task held-out manifest was frozen before outcomes and never accessed (Gate Receipts appendix, Appendix J, Table 2). The instrument-validation window spent 1.19 of 4 GPU-hours, the scaled-adapter window 4.84 of 30, the replication and held-out-panel window 15.22 of 24, the DAgger window 9.06 of 40, the zero-shot window 2.25 of 30, and the second external audit 2.89 of 12, on separate ledgers that are never summed, and the counterfactual and first external audits ran on CPU only.

5.2 ENUMERATION-CONTRACT IDENTITY AUDIT

The identity audit compares random-valid against the exact reference decision by decision on every additive cell of the expanded baseline. On the additive best-first cells of our enumeration contract, where an expansion submits its complete candidate set and heap serials follow the sorted candidate order, random-valid is decision-identical to the exact reference on 48/48 pairs, so success there measures operation validity rather than choice. This is a property of this controller, not additive best-first search in general (Algorithm Invariant Definitions appendix). The BFS and BFWS cells instead register choice, with random-valid at 15/24 and 17/24 against exact at 24/24 (Table 6).

On the same runtime with heap serials assigned in submission order, the same check finds 19/48 divergent pairs, so the check can return either verdict. The pre-registered prediction of divergence on most pairs (more than 24/48) did not hold. Divergent pairs first differ between decisions 2 and 50 (median 6), and on the other 29 pairs every equal-priority tie breaks the same way under both orders (Counterfactual appendix, Appendix S). Under the choice-frontier contract the check diverges on 18/18 pairs (9 tasks, Figure 1).

5.3 CHOICE-FRONTIER CONTRACT AND INSTRUMENT VALIDATION

The choice-frontier contract lets the policy select which runtime-owned frontier state to expand next under a budget of $2\times$ the exact-reference expansion count, and its first measurement on the reused 9-task panel passed the identity gate on 9/9 tasks with random-valid minus exact at -0.644 $[-0.856, -0.422]$ (Comparator-Zoo Receipts appendix, Appendix R, Table 12).

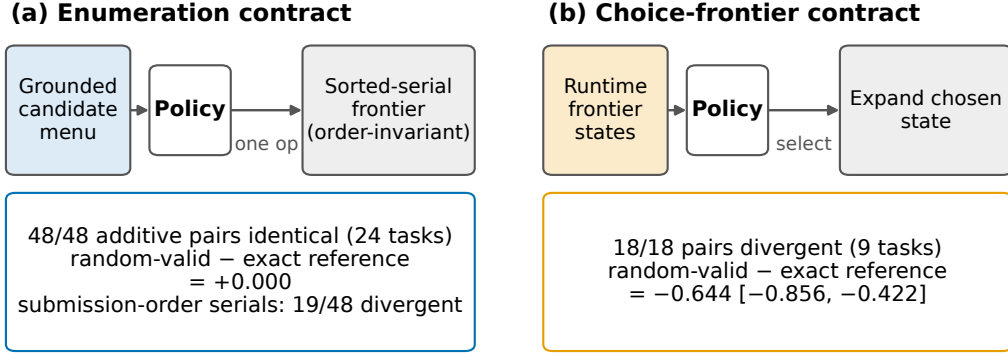


Figure 1: Enumeration versus choice-frontier contract. Under the enumeration contract random-valid is decision-identical to the exact reference on 48/48 additive pairs of the 24-task baseline, and 19/48 pairs diverge when the same runtime assigns heap serials in submission order. Under the choice-frontier contract, on a separate 9-task panel with its own uniform-menu control, the identity gate diverges on 18/18 pairs (9 tasks), and random-valid minus exact is $-0.644 [-0.856, -0.422]$.

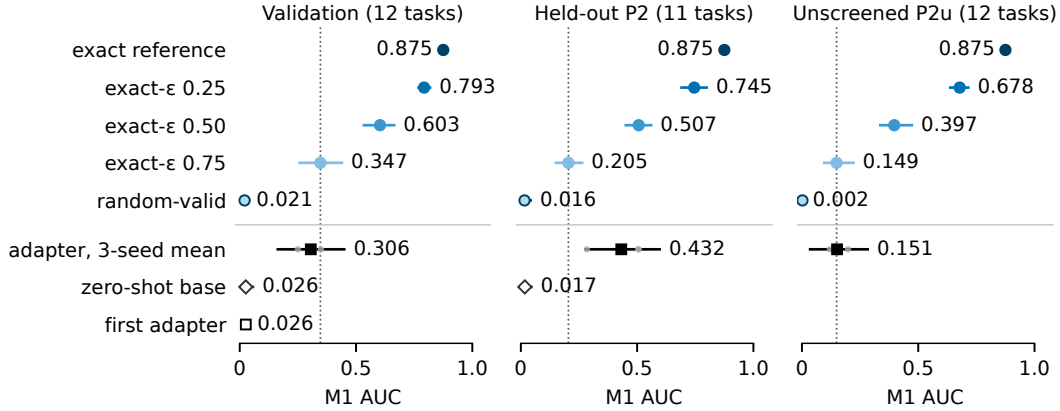


Figure 2: M1 solve-versus-budget AUC with task-clustered 95% intervals on the validation panel, the held-out panel P2, and the unscreened panel P2u, for the five ladder arms and the three-seed adapter mean (per-seed points faint). Ladder rungs read the privileged heap head. The P2u ladder is descriptive, and the first adapter appears on the validation panel only. The zero-shot base model appears on the validation and P2 panels.

On a fresh 12-task panel from 7 domains, the pre-registered ladder test passes. M1 orders the exact reference (0.875) above exact- ϵ selectors that choose uniformly on 25% (0.793), 50% (0.603), and 75% (0.347) of decisions, and random-valid (0.021) lowest, and every adjacent paired task-clustered 95% lower bound is positive (smallest gap $+0.082 [+0.051, +0.111]$). Figure 2 plots each arm, and the Choice-Frontier Validation appendix (Appendix T, Table 17) gives panel construction, every arm’s receipts, and all four paired gaps.

The held-out panel P2 has 11 tasks from 8 domains. P2 applies the validation panel’s generation, cost, and screening rules to fresh seeds, and an amendment made before any evaluation episode, after the first freeze admitted 6 of the 8 required tasks, added seeds 955040–955199 with the rules unchanged. The same ladder test passes on the held-out panel P2, which is pre-registered (smallest adjacent gap $+0.130 [+0.070, +0.190]$), and on the unscreened panel P2u, which is descriptive. The Held-Out Panels appendix (Appendix U.1) gives both panels’ construction and all eight gaps.

The rungs read the privileged heap head, so the ladder validates the ordering of the measurement, not any realisable policy.

5.4 ADAPTER REPLICATION AND HELD-OUT PANELS

Across three training seeds the adapter beats uniform frontier choice on the validation panel, $D_3 = +0.285 [+0.140, +0.433]$, with every seed positive ($+0.328, +0.229, +0.297$), which meets the pre-registered POSITIVE rule, and the result replicates on the held-out panel P2, $D_3 = +0.416 [+0.250, +0.591]$. A later pre-registered replication trained seeds 29 and 71 with the byte-identical recipe (Scaled Adapter appendix, Appendix U). On P2 the per-seed differences are $+0.490, +0.268$, and $+0.490$. The first adapter, re-evaluated on the validation panel, reaches M1 0.026 [0.010, 0.042], at the random-valid level, and the scaled adapter’s last-label rate per seed is 0.205, 0.192, and 0.171 against 0.300 (chance 0.144) for the first adapter on the same panel. The first adapter’s post-hoc M4 intervals on the earlier 9-task panel are in the Counterfactual appendix (Appendix S).

The unscreened panel P2u has 12 tasks from 7 domains. The adapter’s position against the exact- ϵ rungs differs by panel, not separated from the 0.75 rung on the validation panel ($S = -0.041 [-0.187, +0.102]$) and below its 0.50 rung ($-0.298 [-0.468, -0.129]$, descriptive), interval-separated above the 0.75 rung on the held-out panel P2 ($S = +0.227 [+0.053, +0.403]$), and not separated from that rung on the unscreened panel P2u ($+0.002 [-0.145, +0.157]$, descriptive) or pooled over 35 tasks ($+0.058 [-0.040, +0.156]$, descriptive). No equivalence margin was pre-registered for S , so a non-separated S is not equivalence to the rung. The P2 separation rests on 11 task clusters, with 8 positive and 3 negative per-task differences, and percentile intervals can under-cover at this size.

A second realisable policy, the base model prompted zero-shot under the same contract, observation, and cap, is equivalent to uniform choice on the validation panel, $D_3 = +0.005 [-0.025, +0.044]$ within the pre-registered ± 0.05 margin, and inconclusive on P2 ($+0.001 [-0.048, +0.051]$). It falls below the exact- ϵ 0.75 rung and below the adapter on both panels (adapter contrast $-0.280 [-0.425, -0.142]$ and $-0.415 [-0.589, -0.248]$), so the instrument ranks the two realisable policies (Zero-Shot Policy appendix, Appendix U.3).

The random-control screen admits tasks on which near-uniform choice sometimes succeeds, since random-valid must reach the goal in 1 to 9 of its 10 screening episodes. The unscreened panel P2u is drawn from kept candidates outside P2, and all 12 of its tasks had 0/10 random-valid screening goals. Choice still discriminates there, since the ladder passes (descriptive), and the adapter still beats uniform choice, $D_3 = +0.149 [+0.030, +0.283]$, but its gain rests on 4 of 12 tasks, with the other 8 unsolved by both the adapter and random-valid (descriptive). Held-out here means the fresh panel P2, frozen before any evaluation episode from seeds on which no adapter had been evaluated (the seed extension followed a controls-only screen of the first 40 seeds), and the frozen 45-task held-out final evaluation remains unexecuted. Over the three panels the per-task difference is positive on 22 of 35 tasks, zero on 12, and negative on 1 (descriptive). Leaving out one domain at a time over the validation and held-out panels keeps D_3 from $+0.305$ to $+0.395$, with every lower bound at least $+0.185$ (descriptive).

The gain is obtained by imitating the exact reference. The choice-quality claim is development-stage, holds relative to uniform choice and the exact- ϵ ladder on three panels with three training seeds, and carries no planning-ability or generality claim. One on-policy DAgger round, run for one seed on half the planned corpus, does not separate the adapter from the 0.75 rung over the 23 validation and held-out tasks, $S = +0.115 [-0.026, +0.258]$ (DAgger Ablation appendix, Appendix U.2).

5.5 EXTERNAL IDENTITY AUDIT

We audit ScienceWorld (Wang et al., 2022) under its paper’s 100-step budget with five random-valid seeds. On ScienceWorld, a published sequential-choice interface, random-valid diverges from the gold trajectory at the first decision in 1000/1000 pairs, with success 0.915 versus 0 for random-valid over 200 tasks, so success there measures choice. The audit covers 6 of its 30 task types, and none of the seven published interfaces we surveyed uses a complete-set submission contract, so the complete-set identity failure is shown only on our runtime. Four surveyed candidates were not CPU-feasible under their papers’ own budgets. For WebShop and LLM-First Search (Herr et al., 2025) this was a judgement call, and LLM-First Search was the only candidate predicted to saturate (External Identity Audit appendix, Appendix V).

On LLM-First Search (Herr et al., 2025), where the model scores candidates that a search then orders, random-valid diverges from the reference by the third decision in all 400 pairs, yet under the paper’s own token budget it solves all 80 tasks (1.0) against 0.825 for the reference, because a token cap never binds a chooser that spends no tokens. Given only the reference’s own number of expansions, random-valid solves 0.0775, so success there measures choice only under a budget that binds decisions. The reference model is a local open-weight substitute (Qwen3-30B-A3B-Instruct) for the paper’s GPT-4o, whose published success on these configurations is 0.768, and the audit covers 80 tasks from four configurations (LLM-First Search Audit appendix, Appendix V.1).

5.6 ENUMERATION-CONTRACT LEARNED RESULTS AND BOUNDARIES

Primary matrix. The single positive learned-control comparison under the enumeration contract comes from the frozen 15-task text-state BFWS development panel, where one fixed-seed process-SFT run achieved 0.933 success, versus 0.467 for the oracle-assisted random-valid control, 0.0 for the pretrained base, and 1.0 for exact execution, with a paired whole-instance bootstrap 95% interval lower bound 0.2 on the gain (10,000 resamples). On the BFS and the two additive best-first development panels random-valid saturates at 100%, so executable behavior there does not identify learned advantage. The expanded baseline (24 whole problems from 12 domains, four algorithms, three modalities, 576 model episodes independently replayed) gives process SFT 125/288 successes, versus 240/288 for random-valid, 0/288 for the base, and 288/288 for exact execution. Learned success appears only in the additive best-first cells, which the identity audit reads as operation validity, and process SFT scores 0/24 in every modality on the BFS and BFWS rows, where choice registers. Against random-valid, process SFT wins zero of the 72 problem-modality units in any algorithm column. These are tiny-subgroup descriptive bounds over one training seed, and the development-panel BFWS positive does not reappear on this suite (Primary-Matrix appendices, Appendices K and L, Tables 5 and 6).

Secondary evidence. Failure calibration, observation corruption, menu manipulation, the image-only native arms, and the secondary panels (matched modality, generalization, DAgger, successors, curriculum, transfer, second backbone) appear with their receipts in the Results Detail appendices. All are development-stage enumeration-contract evidence outside the choice-quality claim.

Development-stage boundary. All results are development-stage, and the frozen 45-task held-out final primary evaluation is unexecuted, so no final efficacy finding exists. Every enumeration-contract learned cell uses seed 17 except two prospective replication cells ($n=3$, Seed Replication appendix, Appendix W). The full-scope v1 generalization and second-backbone branches were withdrawn after terminal VALID_STOP receipts (closed 2026-09-21). Code and evidence bundles will be provided as anonymous supplementary material.

One panel of nine low-cost tasks (C^* from 3 to 6) serves the corruption suite, the native arms, menu manipulation, the first choice-frontier measurement, and the comparator zoo, and each design step on it could adapt to earlier results. The ladder test and the scaled adapter instead use a fresh panel frozen before evaluation (12 tasks, 7 domains, C^* from 8 to 15), which supplies the fresh non-trivial panel that an instrument-level claim required, and the held-out panel P2 and the unscreened panel P2u were frozen from new seeds before any evaluation episode. The frozen 45-task held-out final evaluation remains unexecuted under the choice-frontier contract.

6 DISCUSSION AND LIMITATIONS

Because the identity check returns both verdicts on one runtime (Section 5.2), random-valid success separates choice from validity only after the check has been run on the interface at hand. The ladder licenses reading a higher M1 as better choice on the validation panel, and the same pre-registered test passes on the held-out panel P2 (Section 5.3). The rungs read the privileged heap head, so the ladder validates the ordering of the measurement, not any realisable policy. Menu-manipulation and zoo results on the earlier contracts are in the Menu-Manipulation Receipts appendix (Appendix P) and the Choice-Frontier and Comparator-Zoo Receipts appendix (Appendix R).

The adapter chooses better than uniform frontier choice on all three panels (Section 5.4), but how far above chance depends on the panel. It is separated above the exact- ϵ 0.75 rung on the held-out panel

P2 and not separated from that rung on the validation panel, the unscreened panel P2u, or pooled (descriptive). Its M1 of 0.151 to 0.432 remains well below the reference’s 0.875. On P2u, whose tasks random-valid never solved in screening, choice still discriminates, but the gain rests on 4 of 12 tasks (descriptive). The result is obtained by imitating the exact reference.

ScienceWorld (Wang et al., 2022) anchors the check on a published interface where it registers choice (Section 5.5). The complete-set identity failure is shown only on our runtime, since none of the seven surveyed published interfaces uses that contract. On the published LLM-First Search interface, random-valid success instead saturates under the paper’s own budget although every decision diverges, a failure the per-decision audit does not detect and a decision-matched budget does. The contribution is therefore the measurement and its two checks, not the exposure of a single flaw.

The portable rule has two checks before random-valid success is read as choice. The per-decision identity audit must find divergent pairs, and the success budget must bind decisions, since a token or time budget that a zero-cost chooser never exhausts lets uniform choice become exhaustive randomized search. When either check fails, success reports operation validity or search coverage rather than choice. Under a contract that passes both, a learned policy’s difference from random-valid should be reported with its position on a privileged exact- ϵ ladder, because the difference alone does not locate choice quality.

Limitations.

- *Ladder position.* Separated above the exact- ϵ 0.75 rung only on the held-out panel, not on the validation panel, the unscreened panel, or pooled.
- *Screening.* P2 is screened like the validation panel, and on the unscreened panel the gain rests on 4 of 12 tasks.
- *Seeds.* Three training seeds, and seed 29 is weakest on every panel (descriptive).
- *Held-out evaluation.* The frozen 45-task held-out final evaluation is unexecuted.
- *Prediction.* The submission-order divergence was weaker than pre-registered.
- *Runtime.* The runtime never chooses or repairs operands, so no result shows unassisted internal search.
- *External audit.* The LLM-First Search reference uses a substituted local model, and the audit covers 80 tasks from four configurations.
- *Second policy.* The second realisable policy is zero-shot, and the trained second backbone was not run.

Code and evidence bundles will be provided as anonymous supplementary material.

7 CONCLUSION

On our runtime a per-decision identity check returns both verdicts. The redesigned choice-frontier measurement passes a pre-registered ladder test, and a scaled adapter beats uniform choice across three seeds, $D_3 = +0.285 [+0.140, +0.433]$, and on the held-out panel P2, $D_3 = +0.416 [+0.250, +0.591]$. It is interval-separated above the exact- ϵ 0.75 rung on the held-out panel ($+0.227 [+0.053, +0.403]$) but not separated from that rung on the validation panel ($-0.041 [-0.187, +0.102]$), the unscreened panel, or pooled over 35 tasks ($+0.058 [-0.040, +0.156]$, descriptive). The choice-quality claim is development-stage, holds relative to uniform choice and the exact- ϵ ladder on three panels with three training seeds, and carries no planning-ability or generality claim. On a published interface, success saturates under a budget that does not bind decisions, so the audit needs both checks.

REPRODUCIBILITY STATEMENT

Reproduction details are in the Trusted Search Runtime, Data and Split Procedures, Receipt Tables, and Statistics appendices. The Choice-Frontier Contract appendix specifies the redesigned contract, and the Counterfactual, Choice-Frontier Validation, Scaled Adapter, and External Identity Audit appendices report the follow-ups, with the Held-Out Panels appendix (Appendix U.1), the DAgger Ablation appendix (Appendix U.2), and the Zero-Shot Policy appendix (Appendix U.3) under the Scaled Adapter appendix, the LLM-First Search Audit appendix (Appendix V.1) under the External

Identity Audit appendix, and ledgers in Table 2. Table 3 indexes the pinned evidence paths. Code and evidence bundles will be provided as anonymous supplementary material.

REFERENCES

- Masataro Asai and Alex Fukunaga. Tiebreaking strategies for A* search: How to explore the final frontier. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 30, 2016. doi: 10.1609/aaai.v30i1.10071. URL <https://ojs.aaai.org/index.php/AAAI/article/view/10071>.
- Shuai Bai, Yuxuan Cai, Ruizhe Chen, et al. Qwen3-VL technical report, 2025.
- Sriram Balasubramanian, Samyadeep Basu, and Soheil Feizi. A closer look at bias and chain-of-thought faithfulness of large (vision) language models. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pp. 13406–13439. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.findings-emnlp.723. URL <https://aclanthology.org/2025.findings-emnlp.723/>.
- Blai Bonet and Héctor Geffner. Planning as heuristic search. *Artificial Intelligence*, 129(1–2):5–33, 2001.
- Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, Girish Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, Nick Ryder, Mikhail Pavlov, Alethea Power, Lukasz Kaiser, Mohammad Bavarian, Clemens Winter, Philippe Tillet, Felipe Petroski Such, Dave Cummings, Matthias Plappert, Fotios Chantzis, Elizabeth Barnes, Ariel Herbert-Voss, William Hebgen Guss, Alex Nichol, Alex Paino, Nikolas Tezak, Jie Tang, Igor Babuschkin, Suchir Balaji, Shantanu Jain, William Saunders, Christopher Hesse, Andrew N. Carr, Jan Leike, Josh Achiam, Vedant Misra, Evan Morikawa, Alec Radford, Matthew Knight, Miles Brundage, Mira Murati, Katie Mayer, Peter Welinder, Bob McGrew, Dario Amodei, Sam McCandlish, Ilya Sutskever, and Wojciech Zaremba. Evaluating large language models trained on code, July 2021.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems, October 2021.
- B. Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979.
- Kanishk Gandhi, Denise Lee, Gabriel Grand, Muxin Liu, Winson Cheng, Archit Sharma, and Noah D. Goodman. Stream of search (SoS): Learning to search in language. In *Conference on Language Modeling (COLM 2024)*, 2024. Spotlight. arXiv:2404.03683.
- Simeng Han, Hailey Schoelkopf, Yilun Zhao, Zhenting Qi, Martin Riddell, Wenfei Zhou, James Coady, David Peng, Yujie Qiao, Luke Benson, Lucy Sun, Alex Wardle-Solano, Hannah Szabo, Ekaterina Zubova, Matthew Burtell, Jonathan Fan, Yixin Liu, Brian Wong, Malcolm Sailor, Ansong Ni, Linyong Nan, Jungo Kasai, Tao Yu, Rui Zhang, Alexander R. Fabbri, Wojciech Kryscinski, Semih Yavuz, Ye Liu, Xi Victoria Lin, Shafiq Joty, Yingbo Zhou, Caiming Xiong, Rex Ying, Arman Cohan, and Dragomir Radev. FOLIO: Natural language reasoning with first-order logic, September 2022.
- Shibo Hao, Yi Gu, Haodi Ma, Joshua Jiahua Hong, Zhen Wang, Daisy Zhe Wang, and Zhiting Hu. Reasoning with language model is planning with world model. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP 2023)*, 2023. arXiv:2305.14992.
- Peter E. Hart, Nils J. Nilsson, and Bertram Raphael. A formal basis for the heuristic determination of minimum cost paths. *IEEE Transactions on Systems Science and Cybernetics*, 4(2):100–107, 1968. doi: 10.1109/TSSC.1968.300136.

- Matthew Hausknecht, Prithviraj Ammanabrolu, Marc-Alexandre Côté, and Xingdi Yuan. Interactive fiction games: A colossal adventure. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2020. doi: 10.1609/aaai.v34i05.6297. URL <https://doi.org/10.1609/aaai.v34i05.6297>.
- Nathan Herr, Tim Rocktäschel, and Roberta Raileanu. LLM-first search: Self-guided exploration of the solution space. arXiv:2506.05213, 2025.
- Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *The Tenth International Conference on Learning Representations (ICLR)*, 2022.
- Yushi Hu, Weijia Shi, Xingyu Fu, Dan Roth, Mari Ostendorf, Luke Zettlemoyer, Noah A. Smith, and Ranjay Krishna. Visual sketchpad: Sketching as a visual chain of thought for multimodal language models. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024. arXiv:2406.09403.
- Sukai Huang, Chenyuan Zhang, Fucui Ke, Zhixi Cai, Naim Rastgoo, Gholamreza Haffari, and Hamid Reza Tofighi. What we talk about when we talk about LLM planning: Evidence for two distinct planning abilities, July 2026.
- Subbarao Kambhampati, Karthik Valmeekam, Lin Guan, Mudit Verma, Kaya Stechly, Siddhant Bhambri, Lucas Saldyt, and Anil Murthy. LLMs can’t plan, but can help planning in LLM-modulo frameworks. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*. PMLR, 2024. arXiv:2402.01817.
- Harsha Kokel, Michael Katz, Kavitha Srinivas, and Shirin Sohrabi. ACPBench Hard: Unrestrained reasoning about action, change, and planning. In *The Fourteenth International Conference on Learning Representations (ICLR 2026)*, 2026.
- Lucas Lehnert, Sainbayar Sukhbaatar, DiJia Su, Qinqing Zheng, Paul Mcvay, Michael Rabbat, and Yuandong Tian. Beyond A*: Better planning with transformers via search dynamics bootstrapping, 2024.
- Chengshu Li, Ruohan Zhang, Josiah Wong, Cem Gokmen, Sanjana Srivastava, Roberto Martín-Martín, Chen Wang, Gabriel Levine, Wensi Ai, Benjamin Martinez, Hang Yin, Michael Lingelbach, Minjune Hwang, Ayano Hiranaka, Sujay Garlanka, Arman Aydin, Sharon Lee, Jiankai Sun, Mona Anvari, Manasi Sharma, Dhruva Bansal, Samuel Hunter, Kyu-Young Kim, Alan Lou, Caleb R. Matthews, Ivan Villa-Renteria, Jerry Huayang Tang, Claire Tang, Fei Xia, Yunzhu Li, Silvio Savarese, Hyowon Gweon, C. Karen Liu, Jiajun Wu, and Li Fei-Fei. BEHAVIOR-1K: A human-centered, embodied AI benchmark with 1,000 everyday activities and realistic simulation, 2024. URL <https://arxiv.org/abs/2403.09227>.
- Chengzu Li, Wenshan Wu, Huanyu Zhang, Yan Xia, Shaoguang Mao, Li Dong, Ivan Vulić, and Furu Wei. Imagine while reasoning in space: Multimodal visualization-of-thought. In *Proceedings of the 42nd International Conference on Machine Learning (ICML 2025)*, 2025. arXiv:2501.07542.
- Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. In *International Conference on Learning Representations (ICLR 2024)*, 2024. arXiv:2305.20050.
- Nir Lipovetzky and Héctor Geffner. Best-first width search: Exploration and exploitation in classical planning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 31, pp. 3590–3596, 2017. doi: 10.1609/aaai.v31i1.11027.
- Olga Loginova, Oleksandr Bezrukov, Ravi Shekhar, and Alexey Kravets. Addressing blind guessing: Calibration of selection bias in multiple-choice question answering by video language models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 3216–3246. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.acl-long.162. URL <https://aclanthology.org/2025.acl-long.162/>.

- Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. Self-refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023. arXiv:2303.17651.
- Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947.
- Maxwell Nye, Anders Johan Andreassen, Guy Gur-Ari, Henryk Michalewski, Jacob Austin, David Bieber, David Dohan, Aitor Lewkowycz, Maarten Bosma, David Luan, Charles Sutton, and Augustus Odena. Show your work: Scratchpads for intermediate computation with language models, 2021. Presented at ICLR 2022.
- Laurent Orseau and Levi H. S. Lelis. Policy-guided heuristic search with guarantees. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, 2021. doi: 10.1609/aaai.v35i14.17469. URL <https://ojs.aaai.org/index.php/AAAI/article/view/17469>.
- Ira Pohl. Heuristic search viewed as path finding in a graph. *Artificial Intelligence*, 1(3–4):193–204, 1970.
- Stéphane Ross, Geoffrey J. Gordon, and J. Andrew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, volume 15 of *Proceedings of Machine Learning Research*, pp. 627–635, 2011.
- Torsten Scholak, Nathan Schucher, and Dzmitry Bahdanau. PICARD: Parsing incrementally for constrained auto-regressive decoding from language models. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP 2021)*, 2021. arXiv:2109.05093.
- Katharina Stein, Daniel Fišer, Jörg Hoffmann, and Alexander Koller. Automating the generation of prompts for LLM-based action choice in PDDL planning. In *Proceedings of the International Conference on Automated Planning and Scheduling (ICAPS)*, volume 35, pp. 250–259, 2025.
- Karthik Valmeekam, Matthew Marquez, Alberto Olmo, Sarath Sreedharan, and Subbarao Kambhampati. PlanBench: An extensible benchmark for evaluating large language models on planning and reasoning about change. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, *Datasets and Benchmarks Track*, 2023a. arXiv:2206.10498.
- Karthik Valmeekam, Matthew Marquez, Sarath Sreedharan, and Subbarao Kambhampati. On the planning abilities of large language models—a critical investigation. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, volume 36, pp. 75993–76005, 2023b.
- Petar Veličković, Rex Ying, Matilde Padova, Raia Hadsell, and Charles Blundell. Neural execution of graph algorithms. In *International Conference on Learning Representations (ICLR 2020)*, 2020. arXiv:1910.10593.
- Ruoyao Wang, Peter Jansen, Marc-Alexandre Côté, and Prithviraj Ammanabrolu. ScienceWorld: Is your agent smarter than a 5th grader? In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP 2022)*, pp. 11279–11298, 2022. URL <https://aclanthology.org/2022.emnlp-main.775/>.
- Weiyun Wang, Zhangwei Gao, Lixin Gu, Hengjun Pu, Long Cui, Xingguang Wei, Zhaoyang Liu, Linglin Jing, Shenglong Ye, Jie Shao, et al. InternVL3.5: Advancing open-source multimodal models in versatility, reasoning, and efficiency, August 2025.
- Brandon T. Willard and Rémi Louf. Efficient guided generation for large language models, 2023.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jing Zhou, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang,

Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report, May 2025.

Shunyu Yao, Rohan Rao, Matthew Hausknecht, and Karthik Narasimhan. Keep CALM and explore: Language models for action generation in text-based games. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 8736–8754, 2020. doi: 10.18653/v1/2020.emnlp-main.704. URL <https://aclanthology.org/2020.emnlp-main.704/>.

Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023. arXiv:2305.10601.

Wojciech Zaremba and Ilya Sutskever. Learning to execute, 2014.

Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. Large language models are not robust multiple choice selectors. In *International Conference on Learning Representations 2024 (ICLR 2024)*, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/54dd9e0cff6d9214e20d97eb2a3bae49-Abstract-Conference.html. arXiv:2309.03882.

Xi Zhu, Zhendong Mao, Chunxiao Liu, Peng Zhang, Bin Wang, and Yongdong Zhang. Overcoming language priors with self-supervised learning for visual question answering. In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence*, pp. 1083–1089, 2020. doi: 10.24963/ijcai.2020/151. URL <https://www.ijcai.org/proceedings/2020/151>.

A OPERATION SCHEMAS

Each declared algorithm defines a vocabulary of operation types τ . A policy emission $u_t = (\tau_t, \theta_t)$ pairs a type from that vocabulary with an operand vector θ_t carrying every exploration-determining argument. The runtime rejects any emission whose type falls outside the declared vocabulary or whose operand vector omits a required field. Every operand that determines exploration must originate from the policy (Section 3).

BFS. The primary operation is node expansion. Its operands name the source state (the current frontier head), the grounded successors selected for insertion, the action producing each successor, and the enqueue position for each. The runtime checks that the named source matches the FIFO frontier head, that each successor results from an applicable grounded action in the source state, that the successor is unvisited, and that each insertion targets the FIFO tail. A goal-test operation names the state to test. The BFS pilot corpus exposes exact grounded successor candidates and full visited membership at each step.

BFWS (complete unpruned). Expansion operands carry the novelty value and goal-count priority of the source state alongside the fields above. Insertion operands carry each candidate’s duplicate membership, novelty, and priority. The runtime checks that the selected source has the declared best priority, that duplicate membership is correctly reported, and that insertions follow the declared ordering. The BFWS corpus exposes candidate duplicate, novelty, priority, insertion, and enqueue facts at every training observation.

Additive best-first. Operands carry the source state’s accumulated path cost g , heuristic estimate h_{add} , and composite value $f = g + w \cdot h_{\text{add}}$ with $w = 3$ for the weighted variant, or $f = h_{\text{add}}$ for the greedy variant. Insertion operands carry per-successor g , h_{add} , and f . The runtime checks that the expanded state has the minimum f in the frontier under the declared tie-breaking rule and that all arithmetic is consistent.

Table 1: Rejection taxonomy from the failure-mechanism calibration. Counts are failures per arm class across 2,910 replay-verified episodes (905 total failures, 0 unclassified). Share is within the arm class.

Arm class	Rejection kind	Failures	Share
Base (447 ep., 447 fail.)	malformed.output	379	84.8%
	invalid.frontier.operation	67	15.0%
	other.invariant.violation	1	0.2%
Learned (906 ep., 407 fail.)	other.invariant.violation	229	56.3%
	malformed.output	72	17.7%
	inapplicable.grounded.action	56	13.8%
	effect	27	6.6%
	schema	16	3.9%
	invalid.frontier.operation	5	1.2%
	source.state.mismatch	1	0.2%
	static.context	1	0.2%
Oracle control (51 fail.)	expansion.budget.exhausted	51	100.0%

Validity preconditions. An emission passes through a fixed sequence of checks. Failure at any step rejects the emission with the corresponding kind and terminates the episode. The checks run in the following order.

1. **Parse.** The raw output must parse into a well-formed Typed Search Operation with a recognized type and complete operand vector. A failure is recorded as `malformed.output`.
2. **Source identity.** The named source state must match the state the algorithm expects. A failure is recorded as `source.state.mismatch`.
3. **Action applicability.** Every grounded action in the operands must be applicable in the source state. A failure is recorded as `inapplicable.grounded.action`.
4. **Successor schema.** Any named successor state must satisfy the domain schema. A failure is recorded as `schema`.
5. **Successor effects.** The named successor must match the declared effects of the applied action. A failure is recorded as `effect`.
6. **Static context.** Referenced predicates, objects, and constants must be consistent with the task instance. A failure is recorded as `static.context`.
7. **Frontier discipline.** The frontier manipulation must respect the declared ordering (FIFO for BFS, priority for BFWS, f -order for additive best-first). A failure is recorded as `invalid.frontier.operation`.
8. **Algorithm invariant.** Any remaining algorithm-specific constraint not covered above. A failure is recorded as `other.invariant.violation`.

The two broad categories are *malformed-output* rejections, where the emission never reaches the operation checker, and *post-parse rejections*, where the emission parses but fails a runtime check. The oracle-assisted random-valid control, which emits only runtime-accepted operations, has a single failure kind: `expansion.budget.exhausted`, recorded when the frozen episode budget runs out before a goal is reached.

Rejection counts. Table 1 gives the counts from the failure-mechanism calibration, which classified all 905 failures across 2,910 replay-verified episodes with zero unclassified. The `effect`, `schema`, and `static.context` kinds arise in successor-prediction episodes, where the model generates a dynamic-state successor.

B ALGORITHM INVARIANT DEFINITIONS

The Trusted Search Runtime checks the following invariants on every operation. Stepwise validity and invariant compliance do not establish termination, completeness, optimality, or episode success (Section 3).

BFS. The frontier is a FIFO queue. Every expansion must select the queue head as its source. Every insertion must target the queue tail. A state already in the visited set must not be re-inserted. This invariant is sufficient for completeness in finite state spaces with unit-cost transitions.

BFWS (complete unpruned). The frontier is a priority queue ordered first by novelty rank (lower is better), then by goal count among states with equal novelty. No generated successor is pruned. A successor whose state already appears in the frontier or visited set is not re-inserted.

Additive best-first, weighted ($w = 3$). The frontier is a priority queue ordered by $f(s) = g(s) + 3 \cdot h_{\text{add}}(s)$, where $g(s)$ is accumulated path cost and $h_{\text{add}}(s)$ is the additive heuristic. Among equal- f states the lower generation serial is expanded first. The expanded state is the heap head. A successor with lower g than an existing frontier entry for the same state reopens that entry with improved cost.

Additive best-first, greedy. The frontier is ordered by $h_{\text{add}}(s)$ alone, with no path-cost component, then by ascending generation serial among ties. The expanded state is the heap head. Reopen and duplicate rules follow the weighted variant. In both additive settings, `BestFirstController.start_expansion` assigns serials before submissions in the deterministic `PDDLStateAuthority.applicable_actions` order (sorted by `GroundedAction.serialize`). The `BestFirstController.apply_operation` heap orders entries by (priority, serial, state identifier, path cost), and `finish_expansion` requires the full candidate set. Thus submission permutations leave frontier order unchanged on the enumeration contract. A runtime assigning serials in submission order would not be submission-order invariant and would not show this identity.

C TRUSTED SEARCH RUNTIME

The runtime provides three guarantees that hold across all declared algorithms.

1. **No-repair property.** The runtime validates, applies, persists to Search Memory, or rejects each emission. It never chooses omitted operands, reorders candidates, silently repairs an invalid emission, or substitutes a default search decision.
2. **Reject-and-terminate.** On the first rejected emission the episode terminates with the rejection kind recorded. There is no retry path.
3. **Replay determinism.** Replaying the raw emissions $u_{1:T}$ against the same task instance and the same initial Search Memory reproduces the explored trace. Any step whose outcome depends on runtime repair would violate this property.

The procedures below specify the per-step check and the episode loop. No `algorithm` package is loaded. The listing uses $\LaTeX$ tabbing.

```

procedure STEP( $u_t, M_{t-1}, A, x$ )
  // Phase 1: Parse the raw emission
  if  $u_t$  does not parse as  $(\tau, \theta)$  with  $\tau \in \text{vocab}(A)$  then
    return REJECT(malformed_output)
  // Phase 2: Validate source identity
  if  $\theta.\text{source} \neq \text{expected}(A, M_{t-1})$  then
    return REJECT(source_state_mismatch)
  // Phase 3: Validate operands against the task and state
  for each grounded action  $a$  named in  $\theta$  do
    if  $a$  is not applicable in  $\theta.\text{source}$  under  $x.D$  then
      return REJECT(inapplicable_grounding_action)
  for each successor state  $s'$  named in  $\theta$  do
    if  $s'$  violates the domain schema then
      return REJECT(schema)
    if  $s' \neq \text{effects}(a, \theta.\text{source})$  then
      return REJECT(effect)
  if any static-context reference in  $\theta$  is inconsistent with  $x$  then
    return REJECT(static_context)

```

```

810 // Phase 4: Check Algorithm Invariant
811 if frontier discipline of  $A$  is violated by  $(\tau, \theta, M_{t-1})$  then
812   return REJECT(invalid.frontier.operation)
813 if any other invariant of  $A$  is violated then
814   return REJECT(other.invariant.violation)
815 // Phase 5: Apply and persist
816  $M_t \leftarrow \text{apply}(\tau, \theta, M_{t-1})$ 
817 return ACCEPT( $M_t$ )
818
819 procedure EPISODE( $x, A, \phi_m, B$ )
820    $M_0 \leftarrow \text{InitSearchMemory}(A, x)$ 
821   for  $t = 1, 2, \dots, B$  do
822      $o_t \leftarrow \phi_m(c_t(x, M_{t-1}))$  // build observation
823      $u_t \leftarrow \pi(o_t)$  // policy emission
824      $r \leftarrow \text{STEP}(u_t, M_{t-1}, A, x)$ 
825     if  $r$  is REJECT(kind) then
826       return ( $t, \text{FAIL}, \text{kind}$ )
827      $M_t \leftarrow r.M$ 
828     if GoalTest( $M_t, x.G$ ) = TRUE then
829       return ( $t, \text{SUCCESS}$ )
830   return ( $B, \text{FAIL}, \text{expansion.budget.exhausted}$ )

```

Each call to STEP consumes one unit of the frozen episode budget B , whether the emission is accepted or rejected. Invalid operations are therefore charged to the budget (Section 4).

D DATA AND SPLIT PROCEDURES

All training and evaluation splits are by whole task instance. No state, decision, or target from an evaluation task appears in any training row. Every released corpus passed no-overlap, no-conflict, live/training-parity, and token-budget audits before use.

BFS pilot corpus. The v6 text-state process corpus holds 25,109 process rows split into 12,994 train and 12,115 dev rows, drawn from 90 replay-verified trusted traces covering 90 distinct tasks. The release assigns splits by semantic task identity, with zero cross-split task overlap. The v8 evaluation panel selects 15 tasks from this corpus’s development set, not its training set. No v8 evaluation-panel task contributes training rows, consistent with the whole-instance rule above. Corpus observations expose the canonical task context, exact grounded successor candidates, and full visited membership. Release audits recorded zero semantic, input, and input-target overlap, zero conflicting identical-input labels, zero teacher-operation rejections, and zero live/corpus mismatches, all within an 8,192-token input cap and a 384-token output allowance.

BFWS corpus. The corpus covers the frozen 105-task development panel with 69,019 replay-verified teacher decisions. Its release contains 69,019 process records, 67,215 operational records, and a process projection of 47,780 train and 21,239 dev rows. Each step’s observation exposes candidate duplicate membership, novelty values, goal-count priorities, insertion positions, and enqueue facts. Audit counters recorded zero overlap, mismatch, rejection, or held-out-access violations.

Evaluation manifests. The BFS pilot uses a frozen 15-task outcome-blind development panel. The BFWS structural gate froze a 105-task development panel and a separate 45-task held-out manifest. The held-out manifest has never been accessed, and no model prediction, training decision, or analysis references it. The expanded baseline evaluates 24 tasks from 12 domains under 4 algorithms, 3 modalities, and 4 arms (1,152 logical bindings, 576 model episodes, all independently replayed). The curriculum stage evaluates 27 unseen tasks. The generalization-v2 protocol admits 25 derived tasks from 93 eligible semantics-preserving variants (uniform $k=5$ cheapest-prefix per family, membership frozen before execution).

Table 2: Combined per-branch gate receipts and ledger allocations. The expanded program spent 68.52 of 336 GPU-h, rounded independently by row. The second-backbone cap reflects a prospective 12.11 GPU-h transfer from recovery reserve. Corruption and seed replication draw on existing branch caps. Follow-up windows have separate ledgers and are never summed into the program total or with each other. The R1-R4 spend covers both stress-test rows.

Branch	Tickets	Terminal state	Frozen scope / replay coverage	GPU-h / cap
Expanded baseline	#116-#118	complete	1,152 bindings, 576 model, 1,152/1,152 replayed	6.53 / 56
Dagger	#78-#84	complete	405 episodes + 81 paired rows, 405/405	10.91 / 48
Successor prediction	#85-#89, #99	complete	90 episodes, 15 cells, 90/90	10.85 / 64
Curriculum by modality	#119	complete	243 model + 324 comparator bindings, 567/567	11.78 / 48
Transfer	#104-#107	complete	39 cells (12 v1, 27 v2), 39/39	3.60 / 24
Second backbone	#101-#103, #123	v1 VALID_STOP, v2 complete	72 model + 72 comparators, 144/144	6.03 / 52.11
Generalization / robustness	#96, #98, #124	v1 VALID_STOP, v2 complete	1,200 episodes (300 model, 900 controls), 1,200/1,200	6.04 / 32
Observation corruption	#126	complete	360/360 new bindings replayed	within generalization
Training-seed replication	#127	complete	534/534 new episodes	12.79 / -
Recovery reserve	-	reduced	-	0.00 / 11.89
Total				68.52 / 336
<i>Follow-up windows (separate ledgers, not summed with the total)</i>				
Native arms	#128, #129	v1 gate history, v2 complete	216/216 new + 72/72 comparators	3.2086 / 12
Stress tests R1/R2/R4	#130	complete	72 menu + 80 variant + 18 cross-cell, 170/170	7.7998 / 36
Second backbone v3 (R3)	#130	complete	72/72 model + 72/72 comparators	same window
Choice frontier	#132	CHOICE_SENSITIVE	144/144 panel, 58 preparation episodes re-derived	1.069 / 12
Comparator zoo	#133	complete	432/432 bindings, 342 new	CPU only
Counterfactual audit, M4 intervals	#134	prediction failed	48 pairs, 288 counts + 96 sequences re-validated	CPU only
Instrument validation	#135	PASS	696/696 CPU + 48/48 model episodes replayed	1.19 / 4
Scaled adapter	#136	POSITIVE	24/24 learned episodes replayed	4.84 / 30
External audit	#137	CHOICE_REGISTERED	1000 pairs over 200 tasks	CPU only
Three-seed replication	#138, #139	POSITIVE	48/48 new learned episodes, 638/638 P2 and 696/696 P2u zoo episodes replayed, and 88/88 P2 plus 96/96 P2u learned episodes	15.22 / 24
held-out panels (v4 window)			24/24 validation-panel and 22/22 P2 episodes replayed	9.06 / 40
Dagger ablation (v5 window)	#140	NOT_SEPARATED	46/46 model episodes replayed	2.25 / 30
Zero-shot second policy (v6 window)	#141	EQUIVALENT (validation), INCONCLUSIVE (P2)	80/80 reference and 400/400 random-valid episodes replayed	2.89 / 12
LLM-First Search audit (external v2)	#142	SATURATED (paper budget)		

E RECEIPT TABLES

Table 2 combines branch-level frozen scope, gate status, coverage, and compute from the synthesis and follow-up ledgers. Five branches completed at v1 scope. Two stopped at frozen v1 cost-admission gates with terminal VALID_STOP receipts and later completed at reduced v2 scope. Every branch reconciles with zero missing evidence. The Tickets column indexes internal ticket numbers.

At the synthesis snapshot, the program recorded 135 attempts (96 succeeded, 36 failed, 3 cutoff). Failed and cutoff attempts retain their hours in the ledger. Every attempt ended before the absolute scheduler cutoff of 2026-09-21T11:55:19Z.

The observation-corruption suite added 2.22 GPU-h to the generalization branch. The training-seed replication addendum (closed 2026-09-21) added 12.79 GPU-h and 534 independently replayed episodes covering two cells. Together these addenda brought the program total to 68.52/336 GPU-h. The Statistics appendix reports the seed-variance results.

Follow-up windows. Each follow-up window keeps its own ledger, and none is added to the program’s 68.52/336 GPU-h. The native-arms window (v1 probes, then v2 training and evaluation) spent 3.2086 of 12 GPU-h. Its 216 new bindings (144 model episodes and 72 CPU random-valid episodes) and 72 reused comparators were all independently replayed. The R1-R4 follow-up window spent 7.7998 of 36 GPU-h across two ledgers. It was resized from 20 GPU-h before any outcome, because its frozen worst-case estimate of 27.04 GPU-h including spend exceeded the initial cap, and realized cost landed at 29% of that estimate. The choice-frontier window spent 1.069 of 12 GPU-h (training 0.81, smoke 0.052, model evaluation 0.208), with corpus derivation and controls on CPU. Its 144 panel episodes were replayed and its 58 preparation episodes re-derived bit-exact. The comparator zoo launched no GPU job. It replayed all 432 zoo bindings, of which 342 are new CPU

Table 3: Artifact index for the follow-up evidence. Each row names a pinned path and the claim class that rests on it.

Pinned path	Claim class
outputs/native-arms/v1/evaluation/ (evaluation.json, analysis.json)	Native-arms clean and corruption success, paired contrasts
outputs/native-arms/v1/identity-audit. json	Identity finding on the enumeration contract
outputs/native-arms/v1/ stress-evaluation.json	Menu manipulation, degraded strains, R2 cross cell, stress replay
outputs/native-arms/v1/r2-decomposition. json	Text-corruption termination modes
outputs/native-arms/v1/budget.json and the R1-R4 window ledgers (v1, v2)	Native-arms and R1-R4 window ledgers
outputs/expanded-study/v1/ second-backbone-v2/evaluation/ outputs/expanded-study/v1/ second-backbone-v3/evaluation/ outputs/choice-frontier/v1/evaluation/ (evaluation, identity-audit, analysis, final-audit)	Second-backbone BFS cell Second-backbone additive replication and its contrasts
outputs/choice-frontier/v1/preparation/ visual-choice-frontier/ (report, audit)	Choice-frontier gate, panel rows, contrasts, re- play
outputs/choice-frontier/v1/budget.json	Choice-frontier corpus derivation
outputs/choice-frontier/o4/zoo/manifest. json	Choice-frontier window ledger
outputs/choice-frontier/ o4/metrics/ (analysis, stored-posthoc-analysis, optimal-costs, native-four-algorithm-reference-costs)	Zoo coverage and replay
docs/experiments/native-arms/ issue-129-closeout.md, issue-130-closeout.md	Exploratory M1–M5 metrics, Holm family, optimal costs, native reference costs
docs/experiments/choice-frontier/ issue-132-closeout.md	Native-arms and stress-test R1-R4 records
docs/experiments/choice-frontier/ metric-adjudication-2026-09-23.md	Choice-frontier outcome and its binding erra- tum
outputs/choice-frontier/v4/seeds/ metrics/analysis.json	Post-close metric adjudication
outputs/choice-frontier/v4/panels/ metrics/analysis.json	Three-seed endpoint, separation, seed variance on the validation panel
outputs/choice-frontier/v5/metrics/ analysis.json	Held-out and unscreened panels, per-panel arms, pooled results, concentration
outputs/choice-frontier/v4/budget.json, outputs/choice-frontier/v5/budget.json	Dagger ablation
outputs/choice-frontier/v6/metrics/ analysis.json	v4 and v5 window ledgers
outputs/choice-frontier/v6/budget.json	Zero-shot second policy
outputs/external-audit/v2/lfs/ identity-audit.json	v6 window ledger
outputs/external-audit/v2/budget.json	LLM-First Search audit
Comparator-zoo ticket (#133) closing comment	External audit v2 ledger
	Zoo numbers table and claim boundary

episodes and 90 are reused $2\times$ random-valid controls. The three-seed replication and the held-out panels shared one window, which spent 15.22 of 24 GPU-h, and the DAgger ablation window spent 9.06 of 40 GPU-h, including 1.79 GPU-h charged for a cancelled collection. The zero-shot window spent 2.25 of 30 GPU-h, and the LLM-First Search audit spent 2.89 of 12 GPU-h on its reference server with no paid API use. Failed attempts retain their hours in every window.

Artifact index. Table 3 maps each pinned follow-up evidence path to the class of claim it supports. Paths are relative to the evidence repository.

F STATISTICS

Default contrast procedure. Contrasts are paired by whole task instance. Paired differences use a bootstrap with 10,000 resamples at seed 1729 and 95% percentile intervals. Under one training seed and panels of 24 or fewer instances, these intervals are descriptive bounds. They do not support broad superiority claims.

The BFWs gate used confidence level 0.95 with 10,000 paired whole-instance resamples (bootstrap seed 1729). Its reported lower bound is the 0.025 quantile, i.e., the lower endpoint of a two-sided 95% percentile interval.

Tiny-strata rule. Strata with fewer than 8 episodes are reported as descriptive only. No intervals are computed over such strata. They are listed explicitly and never silently pooled or dropped.

Transfer tests. Transfer comparisons use McNemar exact tests against the per-example base predictions, with Holm correction across all 36 adapter-by-benchmark comparisons. The minimum adjusted p in the executed transfer branch is 0.8156. None survives correction.

Follow-up bootstrap conventions. The follow-up windows fix their own bootstrap seeds, separate from the main program’s seed 1729 default. The native-arms contrasts and the choice-frontier contrasts use a paired whole-problem bootstrap at seed 61813 with 10,000 resamples and 95% percentile intervals. The second-backbone v3 contrasts (stress test R3) use seed 1729 with 10,000 resamples. The comparator-zoo metrics use a task-cluster bootstrap that draws nine tasks with replacement at seed 133 for 10,000 draws. Both additive cells and all random seeds stay inside their task cluster.

Materiality and non-equivalence. A follow-up contrast is material when its 95% interval excludes zero. The follow-up protocols through the comparator zoo pre-registered no equivalence margin, so a non-material contrast there supports neither superiority nor equivalence. The later choice-frontier protocols pre-register an equivalence margin of ± 0.05 for D and for the DAgger change Δ , and none for the separation S , so a non-separated S is not equivalence to the rung. Wherever the text reports a non-material contrast or a comparison that fails Holm correction, an adjacent sentence states that non-significance is not equivalence.

Tiny strata in the follow-ups. The fewer-than-8 rule also governs discriminative subsets. The choice-frontier learned-minus-random-valid contrast is computed over 18 cells, but it rests entirely on storage, the only panel task where random-valid’s seed frequency lies strictly between 0 and 1 (0.2). A one-task discriminative subset falls under the rule, so that contrast is descriptive. The zoo’s M5 subset, defined from controls only, holds the same single task and is reported descriptively.

Zoo dominance tests. Any positive learned-versus-comparator dominance claim in the comparator zoo uses paired task-cluster bootstrap one-sided p values with Holm step-down correction over five comparators and three metrics, fifteen tests in all. Undefined or uninformative co-solved contrasts prohibit a positive dominance claim and never count as wins. All zoo metrics are exploratory, and those computed on the stored choice-frontier episodes are post-hoc.

Seed-variance addendum. A prospective training-seed replication measured variance for exactly two cells. Seeds 17, 29, and 71 were frozen before any seed-29 or seed-71 artifact existed. All 534 new episodes were independently replayed. Cross-seed aggregation is descriptive ($n = 3$ seeds).

Cell A is the greedy $\times$ multimodal replication cell (`best_first_add_greedy` $\times$ multimodal-state) evaluated on the 24-problem panel. Each seed trained a fresh LoRA adapter with new initialization and dropout randomness on the identical frozen 512-record membership. Table 4 gives the per-seed results.

Cell set B covers DAgger iteration-1 versus the exposure-matched continued-SFT control under BFS across all three modalities. The null replicates, with 0/72 unseen successes for every arm and every seed. The paired contrast (DAgger minus continued SFT) is $+0.000$ $[0.000, 0.000]$ at each seed.

This addendum supersedes the blanket single-training-seed boundary for these two cells only. Every other enumeration-contract cell in the program remains single-seed (seed 17) evidence.

Table 4: Seed-variance replication for Cell A (the greedy $\times$ multimodal replication cell). Contrasts are paired whole-instance bootstrap (24 tasks, 10,000 resamples, seed 1729). The primary contrast (SFT – base) is positive at every seed with no sign flip.

Seed	Successes / 24	SFT – base	SFT – random-valid
17	21/24	+0.875	−0.125
29	21/24	+0.875	−0.125
71	22/24	+0.917	−0.083

G MODALITY ADAPTERS AND OBSERVATION CONTRACT

Three fixed Modality Observation adapters are declared. Each maps a decision context c_t (the canonical task context plus the decision-relevant view of Search Memory M_t) to an observation $o_t = \phi_m(c_t)$ presented to the policy. The adapter is fixed for an entire training or evaluation cell. The adapter contract requires decision sufficiency, meaning that o_t must expose every fact the exact teacher uses to resolve step t under the declared algorithm.

text-state. The state, goal, and search context are serialized as a compact relational representation. Object identities, predicate names, and all Search Memory contents (frontier, visited set, algorithm-specific state) appear as structured text.

visual-state. The state and goal are rendered as a pair of 128×128 pixel images, a rendered state image and a partial-goal constraint image. Object labels are not embedded in the rendered image. Search Memory contents and the grounded candidate list remain in the textual serialization, even in this adapter. This is the published frozen `visual-state` arm that the native arms below are compared against.

multimodal-state. The observation combines the text-state serialization and the visual-state rendered images. The policy receives the relational text and the image pair jointly.

Native image-only arms. Two further adapters executed under the native-arms v2 contract. `visual-nomem-state` keeps only the static task interface and the grounded, unscored candidate menu as text, and carries state facts in the current rendered scene alone. `visual-seq-state` adds up to $K=8$ rendered history frames, each labelled with its absolute path depth. Neither receives textual Search Memory, accepted deltas, g/h /priority scalars, candidate scores, or membership flags.

Text-scaffold caveat. Search Memory contents and the candidate list are serialized as structured text in all three declared adapters, including visual-state. Rendered images carry the planning state and goal visuals, but search bookkeeping (frontier, visited set, novelty counters, candidate list) always travels through the text channel. The observation-corruption suite is consistent with this. Copyability-preserving text shuffling drops success by up to -0.944 (text) and -0.722 (multimodal), while visual corruption costs at most -0.111 with intervals that include zero.

The native arms narrow this caveat to the candidate menu. With the textual bookkeeping and scalars removed, both arms solve 18/18 clean tasks with zero invalid operations, and blanking or degrading every image of the seq arm gives a corrupted-minus-clean contrast of 0.000 [0.000, 0.000]. As a post-hoc interpretation, the grounded unscored menu is the remaining text channel that could carry the signal. Menu manipulation shows that the policies do not reject schema-valid inapplicable menu entries (pick-rate at chance), and on the additive cells submission-order invariance makes any listed candidate a correct emission. The caveat therefore still holds for every adapter that presents the menu, and no executed adapter isolates an image-only decision channel.

Information-loss boundaries. The 128-pixel unlabelled rendered views can lose symbolic identities. The P3 name-compression family in the generalization-v2 suite is information-lossy. Family-wide (24 eligible P3 variants), 24/24 variants are text-lossy, 18/24 are multimodal-lossy, and 0/24 are visual-lossy. Modality matching between training and evaluation ensures identical observation format, not information parity. Static-context and goal fields of the observation contain text in every adapter. The available evidence does not establish semantic parity across adapters or an intrinsic modality effect.

H RETAINED-INFRASTRUCTURE FEASIBILITY NOTES

The terms below describe verified infrastructure components that support corpus generation, evaluation, and evidence production. They constitute feasibility evidence for the runtime and rendering pipeline. They cannot support a Search Process Policy efficacy claim.

Plan pipeline. **Plan Submission** transmits a supplied plan from the project adapter to a planning or rendering backend. **Plan Interpretation** converts the submitted plan into the backend’s ordered action representation. An **Action Sequence** is the canonical identity of a supplied plan as a normalized, ordered sequence of grounded action predicates. Casing and whitespace are not part of its identity.

Rendering pipeline. **Render Production** creates a visualisation output and its rendered image for a planning state. **Render Validation** confirms that the produced visualisation and image satisfy declared structural and semantic checks. **Plan Provenance** links a produced visualisation to the exact supplied plan that the backend interpreted.

Certification and evidence. **Integration Certification** is the finding that Plan Submission, Plan Interpretation, Plan Provenance, Render Production, Render Validation, and required execution-isolation checks all passed. An **Attempt** is one authorized, immutable execution with its own identity and retained outcome, whether successful or not. An **Evidence Bundle** is the self-contained records from an Attempt that allow an independent reproducer to verify its claims offline. The retained infrastructure supplies deterministic compact delta/event evidence with semantic replay while preserving search semantics, serving the replay-determinism requirement of the Trusted Search Runtime.

Historical terms. Planning Certificate, Joint Action-and-Certificate SFT, Adaptive Scaffolding, Support Route, Live Memory, Verified Joint Step, and Route Label are historical CGAS terms. They describe retained infrastructure internals. They cannot support any current Search Process Policy efficacy claim and are not referenced elsewhere in this manuscript.

I CHOICE-FRONTIER CONTRACT AND COMPARATOR SELECTORS

Menu and observation. At each decision the runtime exposes the live frontier as an unscored menu of opaque labels $c_0, \dots, c_K$. Labels are assigned after a seeded Fisher–Yates permutation of the generation-serial member order, with a per-decision seed derived from master seed 51131 and a hash of the member state identifiers, so a label carries no positional information. The observation holds static task-context pages, the unlabelled 128px initial-state scene, one unlabelled 128px scene per frontier state in menu order, and the partial-goal pages. Its text payload carries only the algorithm name, the label list, the representation tag, the schema version, and the frozen legend. A leak check fails closed on any forbidden key marker.

Output and validation. The policy emits canonical JSON with the single key `expand_choice` and a label value. The runtime rejects malformed JSON, a wrong key set, a non-string label, a label not offered by the pending menu, and a label bound to a non-live member. The first rejection ends the episode as `deterministic_invalid_operation`. Other terminations are `goal_reached`, `frontier_exhausted`, `decision_budget_exhausted`, and `expansion_budget_exhausted`.

Expansion and budget. On acceptance the runtime removes the chosen member from the frontier and expands it atomically. Candidate generation, serial assignment, admission, duplicate detection, reopening, and expansion bookkeeping are inherited unchanged from the frozen additive best-first controller, which is not modified. Per-candidate admissions are runtime-internal, so one submitted choice is one decision. Selecting a member that satisfies the goal solves the task without an expansion and without budget charge, so the exact episode takes its reference expansion count plus one decisions. Decision and expansion caps both equal $2 \times$ the reference expansion count.

Corpus derivation. The training corpus was derived on CPU by replaying the frozen exact traces through the new contract and verified expansion-for-expansion against them (58 episodes bit-exact). The frozen matched-modalities membership indexes enumeration-contract decisions. Each such decision maps to the expansion whose candidate set it submitted into, unique (task, expansion) pairs in first-occurrence order form the base set, and a round-robin over the frozen task order tops the set up to 512 training records per algorithm. Diagnostics are the unique expansions of the frozen diagnostic identifiers, with no top-up. The teacher target is the label bound to the heap head. Five fail-closed gates run before training. They check expansion-for-expansion agreement with the stored exact traces, a token budget of measured input plus 384 within 32,768, the text-leak check, image resolution, and identity with the frozen membership. The pinned preparation report records 29 tasks, 1,046 records (512 training and 11 diagnostic per algorithm), 3,544 scenes, and 58 episodes re-derived bit-exact, with outcome PASS. The smoke gate accepted 40/40 model calls.

Training and replay. Each of the two cells trains a LoRA adapter (rank 64, α 128) for 16 updates on its 512 records at seed 17 with the frozen expanded-program recipe. Both training jobs completed all 16 steps and saved final adapters, then failed in post-save summary construction. Their reports were recovered from the primary training-state artifacts with provenance marked, no retraining was needed, and the training audit passes on the recovered reports. At finalize, independent replay re-derives every menu binding, payload, and trusted transition from the stored events, and checks pixel bindings against the retained scenes.

Comparator selectors. Each selector’s choice is mapped back to the menu binding, and every candidate still enters through the additive admission path. Frontier admission, duplicate detection, h_{add} computation, reopening, and serial ordering therefore stay additive for `bfs-order` and novelty-first selection alike. The novelty features of novelty-first selection are positive PDDL state atoms, and its seen-feature sets advance only when a state is expanded, including the initial expansion. Each new episode is re-run against an independently constructed PDDL authority, which checks menus, decisions, admissions, and result dictionaries. Selector episodes replay the model input, menu, actions, admissions, and budget, but they neither generate nor replay image bindings. The optimal costs C^* come from one offline shortest-plan computation per task with replay-verified plans, and the native four-algorithm reference costs are recorded separately from the zoo.

Metric computation. Under M1 an episode solves at multiplier m when its first goal selection occurs within $\lfloor mR_t \rfloor$ decisions with at most $\lfloor mR_t \rfloor$ preceding expansions, where R_t is the additive exact-reference expansion count. Unsolved $2\times$ runs count as failures at every m , and the same prefix truncation applies to every arm. Task and algorithm weights are equal. M2 also has a censored form, in which failures are right-censored at the budget rather than assigned a fictitious expansion count. M3 counts transitions along the selected goal state’s trusted predecessor chain and is undefined, never imputed, for unsolved episodes. M4 measures agreement with the current additive heap head over decisions with at least two live choices, against a chance rate equal to the decision-weighted mean of $1/K$. M5 restricts to tasks where the mean $2\times$ random-valid solve indicator lies strictly between 0 and 1. Because goal selection is the (R_t+1) -th decision, the exact reference fails at $m = 1$ and scores $M1 = 0.875$ by construction, and arms that reach the goal in fewer decisions can exceed it (privileged hadd-greedy 0.882, 0.928, and 0.906).

J RESULTS DETAIL: GATE RECEIPTS AND QUALIFICATION GATES

The expanded study program ran on 2 A100 80GB GPUs under a 336 GPU-hour cap with an absolute scheduler cutoff, and spent 53.52 of those hours (15.9%) at the synthesis snapshot, before the corruption and seed-replication addenda, and 68.5244 after them. Seven branches were admitted. Five completed at v1 scope, and two (second backbone, generalization/robustness) stopped at frozen v1 cost-admission gates with terminal `VALID_STOP` receipts, then completed at reduced v2 scope under new versioned protocols. Table 2 combines frozen scope, status, coverage, and ledger spend. Every executed branch reconciles with zero missing evidence, with 1,152/1,152 baseline episodes independently replayed, 405/405 DAGger episodes, and 90/90 successor-prediction episodes.

Follow-up windows executed after the program on separate GPU ledgers (Table 2) and on CPU only (the comparator zoo, the counterfactual audit, and the first external audit). These windows are

Table 5: Frozen 15-task text-state BFWS development panel, with full-episode budgeted success by condition. The learned-to-best-control gain over random-valid is 0.467 with a paired whole-instance bootstrap 95% interval lower bound 0.2 (10,000 resamples). Single fixed training seed, development stage, held-out manifest not accessed.

Condition	Success (15 tasks)
Pretrained base	0.0
Process SFT	0.933
Random-valid (oracle-assisted control)	0.467
Exact execution	1.0

Table 6: Primary matrix of the expanded 24-task baseline by algorithm and modality. Measures are reported separately and never combined. Columns give full-episode budgeted success out of 24 for each condition, process-SFT invalid operations over decisions (inv/dec), process-SFT expansions, and exact-reference decisions as the budget reference. The pretrained base terminates on its first invalid operation in every cell (24/24 inv/dec, 0 expansions). Random-valid (RV) is an oracle-assisted programmatic control and exact execution is the oracle upper bound, and both are references rather than model abilities.

Algorithm	Modality	Base	SFT	RV	Exact	SFT inv/dec	SFT exp.	Exact dec.
BFS	text	0/24	0/24	15/24	24/24	24/66	15	1969
BFS	visual	0/24	0/24	15/24	24/24	24/69	17	1969
BFS	multimodal	0/24	0/24	15/24	24/24	24/63	15	1969
BFWS	text	0/24	0/24	17/24	24/24	24/24	0	974
BFWS	visual	0/24	0/24	17/24	24/24	24/24	0	974
BFWS	multimodal	0/24	0/24	17/24	24/24	24/24	0	974
Weighted A* ($w=3$)	text	0/24	20/24	24/24	24/24	4/609	179	694
Weighted A* ($w=3$)	visual	0/24	18/24	24/24	24/24	6/499	143	694
Weighted A* ($w=3$)	multimodal	0/24	22/24	24/24	24/24	2/666	196	694
Greedy best-first	text	0/24	22/24	24/24	24/24	2/614	179	673
Greedy best-first	visual	0/24	22/24	24/24	24/24	2/643	189	673
Greedy best-first	multimodal	0/24	21/24	24/24	24/24	3/617	179	673

reported separately and never summed with the main program ledger. The native-arms window spent 3.2086 of 12 GPU-hours, the R1-R4 follow-up window spent 7.7998 of 36, and the choice-frontier window spent 1.069 of 12. The R1-R4 follow-up window was resized from 20 to 36 GPU-hours before any outcome, because its frozen worst-case estimate of 27.04 GPU-hours including spend exceeded the initial cap, and realized cost landed at 29% of that estimate. Each follow-up closed with complete coverage and independent replay (Table 2).

Three earlier gates frame the active algorithm matrix. The bounded BFS pilot closed with a governed VALID_STOP. On a frozen outcome-blind 15-task BFS development panel, final checkpoints from five process-SFT seeds achieved 1.0 invariant-valid success with a 0.0 invalid-operation rate, but the base achieved 0.0 and random-valid achieved 1.0, so the learned policy showed executable BFS behavior with zero gain over the strongest control and downstream work was gated off. The BFWS structural gate then passed on its frozen development panel (Section 5.6). The companion 45-task held-out manifest was frozen before outcomes and has never been accessed. The optimal A* arm over $h_{\max}$ with landmark counting was retired after its own governed VALID_STOP and is superseded by the two additive best-first variants. Its original heuristic-representation estimand remains unidentified. The historical five-seed BFS pilot predates the budget rule, which authorizes one training run at seed 17 per enumeration-contract model and cell (Section 4).

K RESULTS DETAIL: PRIMARY-MATRIX TABLES

Table 5 gives the BFWS development panel and Table 6 the expanded 24-task baseline by algorithm and modality, both summarized in Section 5.6. Measures are reported separately and never combined.

Table 7: Paired whole-problem success contrasts on the expanded baseline (10,000 paired resamples over the 24 problems, seed 1729). Entries give the point contrast with wins/losses/ties over the 72 problem-modality units. Contrasts are descriptive bounds over a tiny subgroup and a single training seed (Efron, 1979).

Contrast	BFS	BFWS	Weighted A* ($w=3$)	Greedy
SFT – base	+0.000 (0/0/72)	+0.000 (0/0/72)	+0.833 (60/0/12)	+0.903 (65/0/7)
SFT – random-valid	−0.625 (0/45/27)	−0.708 (0/51/21)	−0.167 (0/12/60)	−0.097 (0/7/65)
SFT – exact	−1.000 (0/72/0)	−1.000 (0/72/0)	−0.167 (0/12/60)	−0.097 (0/7/65)
Random-valid – exact	−0.375 (0/27/45)	−0.292 (0/21/51)	+0.000 (0/0/72)	+0.000 (0/0/72)

L RESULTS DETAIL: PRIMARY-MATRIX PAIRED CONTRASTS

Paired whole-problem contrasts supply the learned-to-best-control gain, kept separate from the success counts (Table 7). Against the base, process SFT gains only in the additive cells (+0.833 for weighted A*, +0.903 for greedy) and gains nothing under BFS or BFWS. Against random-valid, process SFT loses in every algorithm column and wins zero of the 72 problem-modality units in any column. Random-valid itself ties exact execution in both additive columns, so those cells cannot measure any advantage beyond valid bookkeeping. These contrasts are tiny-subgroup descriptive bounds over 24 problems and one training seed. They do not support modality or algorithm superiority claims beyond the executed panel. The development-panel BFWS positive does not reappear on this suite. The complete six-measure matrix for the frozen final protocol, including per-cell Algorithm Invariant compliance and charged-budget columns, belongs to the held-out evaluation.

M RESULTS DETAIL: SECONDARY PANELS

Modality. The per-modality rows of Table 6 show the same qualitative profile in all three adapters, with zero learned success under BFS and BFWS and learned success below the saturating controls in the additive cells. The earlier matched-modality study, a 3-problem panel on its own ledger that is never pooled with the expanded baseline, gives process-SFT success of 5/12 in text-state, 6/12 in visual-state, and 6/12 in multimodal-state, versus 10/12 for random-valid in each modality, 0/12 for the base, and 12/12 for exact execution. The evidence does not support an intrinsic modality effect. Two caveats bound any such claim. The unlabelled 128px rendered views can lose symbolic identities, and the static and goal pages still contain text in every modality, so semantic parity across adapters is not established.

Generalization and robustness v2. The reduced v2 protocol admitted a uniform cheapest-prefix sample of 5 variants per family over 5 perturbation families (25 of 93 eligible semantics-preserving variants, membership frozen before execution) and evaluated the learned additive-best-first adapters. Learned adapters succeeded on 128/150 episodes versus 0/150 for the pretrained base, where every base episode is a one-call invalid termination (150 decisions total). By family, learned success was 29/30 on object-renaming, 27/30 on name-compression, 27/30 on render-restyle, 26/30 on scale-up, and 19/30 on shifted-init. By modality the split was text 44/50, multimodal 43/50, visual 41/50. By algorithm it was greedy 67/75 and weighted A* 61/75. All 22 learned failures are single-invalid-operation terminations, 11 of them in shifted-init. Controls saturate (random-valid 750/750, exact 150/150) and remain bounds, not learned ability. The lossy name-compression stratum is reported separately and never pooled. All 5 admitted variants are text-lossy, 3 multimodal-lossy, and 0 visual-lossy, with learned success on that stratum of text 9/10, visual 8/10, multimodal 10/10. Family strata hold 5 problems each, so family-level rates are descriptive only. The full-scope v1 generalization branch was withdrawn after its terminal VALID_STOP receipt (closed 2026-09-21) and will not be executed.

Dagger. Two DAgger iterations (Ross et al., 2011) on BFS were compared against an exposure-matched continued-SFT control on 3 development and 24 unseen modality-task pairs (Table 8). On the unseen panel, DAgger and continued SFT each achieve 1/72 invariant-valid successes (paired over whole problems, one DAgger win, one loss, 70 ties), both far below random-valid at 45/72 and exact execution at 72/72, while the original process-SFT checkpoint achieves 0/72. The result is null. DAgger also carries the higher unseen invalid-operation rate of the two iterated arms (0.316

Table 8: DAgger on BFS, with development and unseen success and unseen invalid-operation rates, against the exposure-matched continued-SFT control and the two references.

Arm	Dev success /9	Unseen success /72	Unseen invalid rate
DAgger iteration 2	3/9	1/72	0.316
Continued SFT iteration 2	2/9	1/72	0.175
Original process SFT	1/9	0/72	0.364
Random-valid	6/9	45/72	0.000
Exact reference	9/9	72/72	0.000

Table 9: Zero-shot transfer accuracy on frozen benchmark subsets (FOLIO 200, GSM8K 200, HumanEval 164). Adapter rows follow the transfer ledger names: `bfs`, `iw` (width-based best-first, BFWS), `astar_w3`, and `astar_greedy`, each in text (t), visual (v), and multimodal (m) training cells. No comparison against the base survives Holm correction (0/36, minimum adjusted p 0.8156).

Adapter	FOLIO	GSM8K	HumanEval
Base	0.600	0.955	0.780
<code>bfs t/v/m</code>	0.600 / 0.605 / 0.605	0.950 / 0.945 / 0.965	0.793 / 0.799 / 0.780
<code>iw t/v/m</code>	0.660 / 0.645 / 0.645	0.960 / 0.965 / 0.960	0.768 / 0.762 / 0.756
<code>astar_w3 t/v/m</code>	0.590 / 0.575 / 0.590	0.950 / 0.965 / 0.945	0.817 / 0.799 / 0.774
<code>astar_greedy t/v/m</code>	0.590 / 0.590 / 0.590	0.950 / 0.955 / 0.950	0.799 / 0.811 / 0.793

versus 0.175). Final cumulative unique training corrections were text 201, visual 203, multimodal 181, accumulated over 12 training cells of 512 records and 16 optimizer updates per cell-iteration.

Model-generated successors. Replacing trusted successors with model-generated dynamic-state successors collapses goal success from 45/45 to 1/45 on the evaluated panel, with zero trusted-state substitutions in the trusted arm. The single success is the one multimodal unseen cell. The trained successor model is exactly correct on 65/109 retained raw predictions, with 93 schema-valid and 66 effect-valid, and downstream failure kinds are dominated by exact state-identity and effect failures (27 effect, 16 schema, 1 static-context) rather than schema failures. Collection-time verification of the untrained exact-successor accepted 119/1,536 predictions (text 43, visual 52, multimodal 24). Coverage is partial by design. The frozen outcome-blind admission accepted a 12-task fallback panel after the full 24-task unseen panel was priced at 147.66 GPU-hours against the 64 GPU-hour branch cap, so full unseen coverage remains missing.

Curriculum by modality. Nine training cells (three modalities under staged, shuffled, and mixed ordering, all under greedy additive best-first and trained fresh from the same base on identical record sets) were evaluated on 27 unseen whole problems. Successes were 24/24/25 (staged/shuffled/mixed) in text-state, 24/27/26 in visual-state, and 25/27/27 in multimodal-state. A sequential-SFT comparator reached 24/27, 25/27, and 24/27 in text, visual, and multimodal cells respectively, the base stayed at 0/27, and random-valid and exact execution saturated at 27/27 in every modality. All four paired interaction intervals include zero: $-0.111 [-0.259, +0.037]$ and $-0.074 [-0.259, +0.074]$ for the visual and multimodal staged-minus-shuffled contrasts against text, and $-0.074 [-0.259, +0.111]$ and $-0.037 [-0.185, +0.111]$ for the corresponding mixed-minus-shuffled contrasts. No material ordering-by-modality interaction emerged. Because both references saturate and the base sits at floor, ceiling and floor effects bound the observable differences.

Transfer. All 12 verified planning adapters plus the base were evaluated zero-shot on frozen subsets of FOLIO (Han et al., 2022) (200 items), GSM8K (Cobbe et al., 2021) (200), and HumanEval (Chen et al., 2021) (164). Table 9 gives the per-adapter accuracies. The paired verdict uses McNemar exact comparisons (McNemar, 1947) of each adapter against the per-example base predictions, Holm-corrected across all 36 comparisons (Holm, 1979). None survives correction. The minimum adjusted p is 0.8156, and the smallest raw p is 0.0227 (FOLIO with the text width-based adapter, at +12 examples). Leakage screening over the frozen 8-gram window finds 7/564 benchmark items sharing only trivial numeric 8-grams with the planning corpus and zero content overlap. The transfer result is a null.

Second backbone. Two reduced-scope cells trained and evaluated InternVL3.5-8B (Wang et al., 2025). Each used a frozen 12-task key-cell panel selected by lowest reference decision count before

any model outcome, with one seed-17 training run per modality cell. The earlier cell (protocol v2) ran BFS, and neither arm reached a goal. The pretrained base scored 0/36 at 1 decision per episode (36 total), process SFT scored 0/36 at 1 to 4 decisions (59 total), and the reused BFS controls scored random-valid 54/72 (18/24 per modality, oracle-assisted) and exact 72/72. The replication cell (protocol v3, stress test R3) ran greedy additive best-first with the same backbone, recipe, and panel shape. Process SFT succeeds on 11/12 text-state, 10/12 visual-state, and 10/12 multimodal-state problems (31/36), while the base scores 0/36 with one invalid call per episode. Process-SFT invalid-operation rates are 0.00546, 0.01220, and 0.01258, and decision usage is 183, 164, and 159 of a 390-call allowance (text, visual, multimodal). Random-valid and exact reference both score 24/24 per modality. These control counts hold two entries per panel task, and both entries of a task come from its one reused, hash-pinned greedy baseline control episode per condition and modality. Paired whole-problem contrasts (seed 1729, 10,000 resamples) give SFT minus base +0.917 [+0.750, +1.000] in text-state and +0.833 [+0.583, +1.000] in both visual-state and multimodal-state. SFT minus random-valid is -0.083 [-0.250, +0.000], -0.167 [-0.417, +0.000], and -0.167 [-0.417, +0.000] in the same order. Coverage is complete, with 72/72 model episodes and 72/72 comparators replayed. Learned success under the enumeration contract appears only in the additive cells, where the identity audit shows no decision to make (Section 5.2), and the InternVL v3 cell replicates that contract fluency (31/36 versus 0/36 base, below random-valid 24/24 per modality). The earlier total null on this backbone is consistent with an algorithm (choice-requirement) difference rather than a backbone null. The cross-backbone contrast against the pinned Qwen3-VL process-SFT rate is descriptive rather than paired, because the InternVL panel is the key-cell subset while the Qwen rate is the pinned full-panel aggregate. It is +0.000 [-0.167, +0.083] in text-state, -0.083 [-0.333, +0.083] in visual-state, and -0.042 [-0.292, +0.125] in multimodal-state. InternVL3.5-8B shares the Qwen3-8B LLM family (Yang et al., 2025), so the comparison differs in the vision tower, connector, image tokenization, and multimodal recipe, and it is reported as an architecture-difference limitation. The full-scope v1 second-backbone branch was withdrawn after its terminal VALID_STOP receipt (closed 2026-09-21) and will not be executed, and its gate is retained as history.

N RESULTS DETAIL: FAILURE-MECHANISM CALIBRATION

A frozen CPU-only calibration mined 2,910 unique replay-verified episodes across the five executed model branches and classified all 905 failures with zero unclassified. The protocol was frozen before any outcomes were computed, and the analysis is descriptive. Three arm-level profiles emerge. The base policy fails in 447 of 447 episodes, always on the first decision. Of these, 379 emit a malformed output that never reaches the operation checker, and 67 issue rejected BFS frontier operations. Learned arms fail in 407 of 906 episodes, and the dominant learned failure kind is `other_invariant_violation` with 229 occurrences (56.3% of learned failures), namely parsed, source-correct, applicable operations that the runtime rejects because they violate the declared search invariant. The remaining learned failures are 72 malformed outputs, 56 inapplicable grounded actions, 27 effect failures, 16 schema failures, and smaller kinds. First failures cluster early, at decision indices 0, 1, and 2 with counts 137, 88, and 46. The oracle control fails 51 times, all late and all by expansion-budget exhaustion (buckets 5–8, 9–16, 17–32, and 33+ hold 9, 9, 3, and 30). Table 1 gives the rejection taxonomy.

Every baseline process-SFT episode under BFS or BFWS fails (144/144). The DAgger-iteration BFS arms fail 155/162, while the learned additive best-first families fail 56 of 546 episodes (10.3%). This pattern matches the primary matrix (Section 5.6). The calibration is descriptive. After process supervision, most learned failures (229/407) are parsed, source-correct operations rejected by the search invariant, and 72/407 remain malformed outputs. Read with the horizon profile, the learned policy fails early on invariant grounds, whereas the oracle control, which always emits valid operations, fails only when its expansion budget runs out late in the episode.

O RESULTS DETAIL: OBSERVATION-CORRUPTION CHANNEL TESTS

The observation-corruption suite evaluated frozen checkpoints, with no retraining, on a prospectively frozen set of $k=9$ admitted tasks under four deterministic corruption families applied at observation time. Clean learned success on these tasks is 0.917 (text-state 0.944, visual-state

Table 10: Observation corruption on frozen checkpoints ($k=9$ admitted tasks, paired whole-problem deltas against clean runs, strata below 8 descriptive-only under the frozen rule). “Material” is the frozen protocol’s interval-based verdict.

Corruption	Arm	Corrupted	Clean	Δ	95% interval	Material
Text-masked	text	0.000	0.944	-0.944	$[-1.000, -0.833]$	yes
Text-masked	multimodal	0.000	0.889	-0.889	$[-1.000, -0.722]$	yes
Text-shuffled	text	0.000	0.944	-0.944	$[-1.000, -0.833]$	yes
Text-shuffled	multimodal	0.167	0.889	-0.722	$[-0.889, -0.500]$	yes
Visual-blank	visual	0.833	0.889	-0.056	$[-0.167, +0.000]$	no
Visual-blank	multimodal	0.778	0.889	-0.111	$[-0.278, +0.000]$	no
Visual-degraded	visual	0.833	0.889	-0.056	$[-0.167, +0.000]$	no
Visual-degraded	multimodal	0.778	0.889	-0.111	$[-0.278, +0.000]$	no

0.889, multimodal-state 0.889, with 9 tasks $\times$ 2 algorithms per cell). Table 10 gives the paired whole-problem deltas. Copyability-preserving text shuffling lowers success by up to -0.944 $[-1.000, -0.833]$, and the multimodal arm retains at most 0.167 success under shuffled text. Masking instead destroys the output contract. Visual-channel corruption is not material. Blanking or degrading the rendered state costs at most -0.111 $[-0.278, +0.000]$, and every visual interval includes zero. The multimodal channel-isolation contrast, paired by task and algorithm ($n=18$), is -0.806 for text-channel corruption versus -0.111 for visual-channel corruption, giving text minus visual -0.694 with 95% interval $[-0.889, -0.472]$, which is material under the frozen rule. This contrast pools the output-contract-destruction (masked) regime. Controls behave as the contract predicts. The corrupted base stays at floor (0/144, all first-decision invalid), random-valid succeeds 72/72 on the new frozen seeds, so per the saturation rule no structural-advantage claim is made, and exact execution remains at 18/18.

A decomposition of the 72 published text-corruption episodes (stress test R2) separates two regimes that the success deltas pool together. Under text masking, every episode of the text-state and multimodal-state cells is all-invalid (18/18 in each, 18 invalid operations over 18 decisions). Each emitted operation fails the schema, so the policy never enters the search. This is output-contract destruction, not information loss. Copyability-preserving text shuffling behaves differently, because valid operations keep flowing while the search fails. In the text-state cell 16/18 episodes are partial-valid, 2/18 are all-invalid, and none reaches the goal (18 invalid operations over 49 decisions). In the multimodal-state cell 15/18 are partial-valid, none is all-invalid, and 3/18 reach the goal (15 invalid operations over 105 decisions). This partial-valid pattern is the information-loss signature. A new cross cell applies text masking to the frozen visual-state arm, whose nominal state channel is the image, and it scores 0/18 with every episode all-invalid. Masking the text scaffold destroys the output contract even when images are the nominal state channel.

The language-channel claim survives only in its shuffled form. The masked catastrophe in Table 10 is mostly a loss of the copyable output format rather than a loss of state information. Shuffling the text leaves operations copyable and still stops the search, so under the frozen observation contract the textual serialization carries decision-relevant information for these adapters. Removing or degrading the visual channel barely moves success, including on the visual-state arm whose only rendered-semantics channel was corrupted. Sections P and Q probe the grounded candidate menu directly. This is a channel finding about the trained controller under the declared adapters, not evidence about intrinsic modality value.

P RESULTS DETAIL: MENU-MANIPULATION RECEIPTS

Menu manipulation (stress test R1) probes how the trained policy responds to changes in the grounded menu. It runs on the frozen image-only adapters of both native arms (the Image-Only Native Arms appendix, Appendix Q), with no retraining, under two families. Menu permutation reorders the candidate menu. Distractor injection adds $k=3$ schema-valid but inapplicable actions to it. Table 11 reports the 72 episodes, all independently replayed within the 170/170 stress-evaluation episodes.

On the frozen native adapters (stress test R1, 72 replayed episodes), menu permutation leaves 36/36 successes and 0/512 invalid operations, which is uninformative about order under the submission-order invariance of Section 5.2. Injecting $k=3$ schema-valid but inapplicable actions collapses suc-

Table 11: Menu manipulation on the frozen native adapters (both arms, two additive algorithms). Invalid operations are counted over emitted operations. The distractor pick-rate is counted over decisions with injected distractors.

Family	Success	Invalid ops	Distractor picks
Menu permutation	36/36	0/512	n/a
Distractor injection	0/36	36/60	36/60 (0.6)

cess to 0/36, with 36/60 invalid operations and a distractor pick-rate of 0.6 against a uniform-picking expectation of 0.592 to 0.625, and every episode ends on a distractor pick, 20/36 on the first decision. The receipt therefore shows that the policy does not reject schema-valid inapplicable menu entries (pick-rate at chance), and these out-of-distribution distractors never appeared in training menus. Content-reading and position habits both predict a pick-rate near 0.6, so this receipt does not separate them, and the discriminating fixed-versus-random-position test remains unrun.

Q RESULTS DETAIL: IMAGE-ONLY NATIVE ARMS

The native-arms follow-up (contract v2) trained two image-only observation arms. In `visual-nomem-state` (nomem) and `visual-seq-state` (seq), state facts arrive only as rendered images, and seq adds a window of $K=8$ history frames. Neither arm receives textual search memory, g/h /priority scalars, candidate scores, or membership flags. The grounded, unscored candidate menu is still presented. The v1 admission gate measured pretrained-base format compliance and is retained as history. The v2 amendment, frozen before any trained-arm outcome, replaced it with a pre-registered post-training smoke gate, which both arms passed with 62/62 schema-valid grounded calls against a threshold of 0.5. Four adapters (two arms by two additive best-first algorithms) used the frozen recipe with 512 records, 16 updates, and seed 17. All 216/216 new bindings were recorded and independently replayed, together with 72/72 reused replay-verified comparators. The window spent 3.2086 of 12 GPU-hours.

On the clean panel, both arms nominally succeed on 18/18 with zero invalid operations, against 16/18 for the published frozen `visual-state` arm. The pretrained base scores 0/36 under both new contracts and exact reference scores 18/18. Random-valid is at ceiling on its five frozen seeds, so under the saturation rule no structural-advantage claim is made. Paired whole-problem contrasts ($n=18$, bootstrap seed 61813, 10,000 resamples, 95%) give nomem minus `visual-state` +0.111 [0.000, +0.278] and seq minus `visual-state` +0.111 [0.000, +0.278], neither material. The history window changes nothing on these tasks (seq minus nomem 0.000 [0.000, 0.000]). Learned minus pretrained base is +1.000 [+1.000, +1.000] for each arm, which is material.

The pre-registered corruption payoff is a null. Under seq the images are the only state and history channel, yet blanking every attached image gives a seq corrupted-minus-clean contrast of 0.000 [0.000, 0.000], and degrading every image gives 0.000 [0.000, 0.000]. The images-only hypothesis is therefore answered in the negative on this panel. As a post-hoc interpretation, the grounded unscored candidate menu is the remaining channel that could carry the signal, since neither the rendered scenes nor the removed scalar scaffold is necessary once the menu is present. On these additive cells the identity audit (Section 5.2) makes any listed candidate a correct emission, so the null is equally consistent with a policy that uses no state information.

The degraded-strain test (stress test R4) evaluated the same frozen native adapters on the two generalization families where the full-scaffold adapters had degraded. The learned adapters succeed on 10/10 shifted-init and 10/10 scale-up episodes for both arms, and the pretrained base scores 0/20 on each strain (0/40 in total). The published full-scaffold rates on the same variants are 19/30 for shifted-init and 26/30 for scale-up (Secondary Panels appendix, Appendix M). The two rates come from different episode sets and are not paired. The reduced contract sits at ceiling where the full scaffold degraded. A post-hoc, unpaired reading, with 10 visual-only episodes per arm against 30 full-scaffold episodes pooled over three modalities, is consistent with the reduced contract removing scalar operands whose emission the full-scaffold adapters failed, and it does not measure whether that text carried signal, since any listed candidate succeeds on these cells.

Table 12: First measurement under the choice-frontier contract on the frozen 9-task panel, with 18 cells (9 tasks $\times$ 2 additive algorithms as repeated measures) and 144 episodes independently replayed. Exact reference solves by construction of the budget and is the imitation target. Not comparable with enumeration-contract rows.

Condition	Goal rate	Failures	Invalid ops
Exact reference	18/18	none	0
Learned adapter	7/18	11 budget exhausted	0
Pretrained base	0/18	18 invalid at first call	18
Random-valid (5 frozen seeds)	32/90	58 budget exhausted	0

Table 13: Exploratory pre-registered metric set at the frozen shared $2\times$ cap (9 panel tasks $\times$ 2 additive settings). M1 is the task-clustered solve-versus-budget AUC over $m \in \{1, 1.25, 1.5, 1.75, 2\}$ (9-task bootstrap, seed 133, 10,000 draws). M2 is the solved-only task-weighted geometric expansion overhead ρ and M3 the solved-only task-weighted cost ratio κ , both selection-conditioned. The $3\times$ and $4\times$ columns are additional CPU controls, not M1. Selectors read privileged information that the model does not see.

Arm	M1 AUC [95% CI]	$2\times$ solved	$3\times$	$4\times$	M2 ρ	M3 κ
Learned adapter	.1042 [.0278, .2014]	7/18	n/a	n/a	1.432	1.025
Pretrained base	0 [0, 0]	0/18	n/a	n/a	undefined	undefined
Random-valid	.1389 [.0250, .2722]	32/90	42/90	50/90	1.407	1.000
Exact reference	.8472 [.7917, .8750]	18/18	n/a	n/a	1.000	1.000
bfs-order	.0694 [0, .1528]	6/18	8/18	10/18	1.613	1.000
Novelty-first selection	.1250 [0, .2778]	6/18	8/18	12/18	1.452	1.000
worst-first	.0556 [0, .1389]	4/18	6/18	8/18	1.549	1.050

R RESULTS DETAIL: CHOICE-FRONTIER AND COMPARATOR-ZOO RECEIPTS

The training corpus was derived on CPU by replaying the frozen exact traces through the new contract and verified expansion-for-expansion against them (58 episodes bit-exact). It covers 29 training tasks and holds 1,046 records (512 training plus 11 diagnostic per algorithm) with 3,544 scenes. Every corpus episode carries at least one decision whose menu offers two or more states (up to 315 such decisions in one episode, menus of up to 222 states). Two adapters, one per additive algorithm, were trained with the frozen recipe at seed 17 (16 updates, 512 records each), and the smoke gate accepted 40/40 model calls. All 144/144 panel episodes were independently replayed with complete coverage. The window spent 1.069 of 12 GPU-hours (training 0.81, smoke 0.052, model evaluation 0.208), and corpus and controls ran on CPU.

Paired bootstrap contrasts (seed 61813, 10,000 resamples) give random-valid minus exact reference $-0.644 [-0.856, -0.422]$, material, so the controls separate under this contract. The learned adapter minus pretrained base is $+0.389 [+0.167, +0.611]$, material, with zero invalid operations, a contract-fluency gain. The learned adapter minus random-valid is $+0.033 [-0.033, +0.133]$, not material. The 18 cells are 9 tasks with two algorithms as repeated measures, and that last contrast rests entirely on one task (storage), the only one where random-valid’s seed frequency lies strictly between 0 and 1. That falls below the frozen rule that strata under eight cells are descriptive only, so the contrast is descriptive. The evidence supports no learning-to-plan claim for the adapter, and no equivalence claim with random-valid, since no equivalence margin was pre-registered.

The control expansion sequences coincide across the two algorithms on every panel task. Post-hoc task-clustered intervals with the same seed and resample count are $[0.000, +0.100]$ for learned minus random-valid and $[-0.889, -0.333]$ for random-valid minus exact reference.

Comparator zoo. The CPU-only comparator zoo evaluates this measurement against three deterministic selectors at the same $2\times$ cap (bfs-order, novelty-first selection, and negative control worst-first). Its metric set was pre-registered before comparator episodes, but stored choice-frontier episodes enter post-hoc. The selectors act on the additive frontier, not native BFS or BFWs, and read privileged scalars or PDDL atoms while the model sees only scenes, so no information-matched learned-versus-rule claim follows.

On M1 the learned adapter falls between bfs-order and random-valid and below novelty-first selection (Table 13, Figure 3). No comparator between random-valid and exact exists on this panel, so the zoo bounds the adapter only against weaker-than-random rules and cannot calibrate choice quality. Post-hoc M4 finds 17.4% heap-head agreement against 26.9% chance and 54.5% last-menu-label emissions, and the task-clustered interval of the agreement gap includes zero (Counterfactual

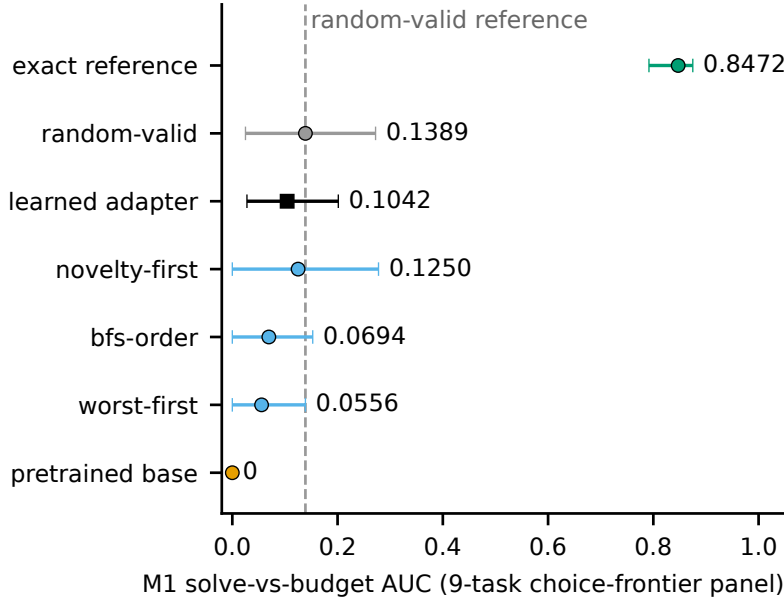


Figure 3: Solve-versus-budget AUC (M1) on the frozen 9-task choice-frontier panel at multipliers 1 to 2. Exact reference .8472, random-valid uniform-choice reference .1389, learned adapter .1042 [.0278, .2014], novelty-first .1250, bfs-order .0694, worst-first .0556, pretrained base 0. Intervals of all non-exact arms overlap and the Holm family licenses no ordering. Under the enumeration contract, the BFS/BFWS random-valid rows (15/24 and 17/24 versus exact 24/24) already registered choice.

appendix, Appendix S), with no last-label comparator and no equivalence claim. Across the fifteen-test Holm family no comparison supports positive learned Pareto dominance, with three co-solved tests undefined. Non-significance is neither equivalence nor a generalized negative capability claim, and the one-task M5 subset is descriptive only.

Zoo receipts. Random-valid is the zoo’s uniform-choice reference and exact reference the imitation target. Random-valid and the three selectors also ran at $3\times$ and $4\times$ caps, outside the primary metric. Because the CPU selectors operate on the additive frontier, novelty-first selection does not reproduce BFWS pruning. Selector episodes replay the frozen model input, randomized menu, actions, admission, and budget, but do not generate or replay image view bindings. All 432/432 zoo bindings were observed and independently replayed, with 342 new CPU episodes, 90 original $2\times$ random-valid controls reused, and none missing. The nine unit-cost optimal costs C^* come from one offline planner computation per task with replay-verified plans: storage 3; elevators, ferry, and 15puzzle-compact 4; depot, towers-of-hanoi, and visitall 5; blocksworld and 15puzzle-expanded 6.

The selectors are `bfs-order` (minimum frontier g , then earliest serial), novelty-first selection (lowest width-1 then width-2 novelty over positive PDDL atoms), and `worst-first` as a negative control.

Exploratory post-hoc M4 on the stored episodes shows the learned adapter selecting the teacher’s heap-head state on 23/132 multi-option decisions (17.4%) against an on-policy chance rate of 26.9%, and emitting the last menu label on 72/132 (54.5%). Decisions cluster within 9 tasks, and the task-clustered intervals in the Counterfactual appendix (Appendix S) show that the agreement gap does not survive that clustering while the last-label excess does. The post-hoc choice-frontier diagnostic records a first-label rate of 2.3% and a teacher-corpus last-label rate of 12.0% (chance 10.4%, 978 records). Menus are permuted per decision with a seed, and no last-label comparator was run, so these post-hoc rates do not establish a position habit. M5 restricts to tasks where random-valid solves at a $2\times$ rate strictly between 0 and 1. That subset holds one task (storage-compact-919000, random-valid rate .2), so M5 is descriptive only. On it the learned adapter’s M1 AUC is .3125, against .075 for random-valid and .625 for exact reference.

Table 14: Random-valid against exact reference on the 48 additive pairs under the two heap-serial rules.

Heap-serial rule	Identical	Divergent
Sorted candidate order (executed)	48	0
Submission order (counterfactual)	29	19

Table 15: Post-hoc M4 diagnostics for the first adapter on the earlier 9-task panel.

Quantity	Pooled	Task-clustered [95% CI]
Heap-head agreement minus chance	-0.095	-0.069 [-0.161, +0.032]
Last-label rate minus chance	+0.276	+0.277 [+0.126, +0.402]

The choice-frontier contrasts use percentile 95% intervals with the frozen bootstrap resampling cells. Random-valid’s seed frequency on storage is 0.2. The final audit returns CHOICE_SENSITIVE and passes over 144 replayed episodes.

S RESULTS DETAIL: SUBMISSION-ORDER COUNTERFACTUAL AND M4 INTERVALS

A CPU-only follow-up with a protocol committed before its outputs re-ran the identity check on the same 48 additive pairs (24 tasks $\times$ 2 algorithms, seed 17). The executed rule assigns heap serials at expansion start from the sorted candidate order, and the counterfactual assigns them at application time in submission order. The audit first re-validated 288 stored episode counts and 96 raw-output sequences with no mismatch.

The stored verdict reads “PREDICTION_FAILED: see preregistered_prediction”. The executed rule stays identical as predicted, but 19/48 divergent pairs fall short of the pre-registered threshold of more than 24/48. First divergence falls between decisions 2 and 50 (median 6), and both algorithms on one ferry task flip the goal outcome. The 48/48 identity is therefore a property of the executed serial rule, not of the enumeration contract alone.

A post-hoc analysis recomputed the first adapter’s M4 diagnostics over its 18 stored episodes on the earlier panel (9 task clusters, 10,000 resamples, seed 133). Pooled over 132 multi-option decisions, heap-head agreement is 0.174 against a chance rate of 0.269, and the last-label rate is 0.545. The agreement gap’s interval includes zero while the last-label excess excludes it (Table 15), and M4 keeps its diagnostic, non-confirmatory status.

T RESULTS DETAIL: CHOICE-FRONTIER VALIDATION PANEL AND LADDER

The ladder protocol was committed before task generation and the panel membership before evaluation. Of 240 generated candidates (12 snapshots, seeds 935000 to 935019), 127 met the cost filter ($8 \leq C^* \leq 20$) and 34 passed controls-only screening, which admits tasks on which random-valid is discriminative. The freeze rule sorts admitted tasks by domain and seed, keeps at most two per domain, and takes the first 12 (Table 16).

All 696/696 CPU episodes and the 48 first-adapter model episodes were independently replayed. PASS requires all four adjacent paired task-cluster lower bounds to be positive, with arms averaged over seeds within each task and algorithm, then over algorithms. The gaps are +0.082 [+0.051, +0.111] (exact reference minus 0.25), +0.190 [+0.126, +0.252] (0.25 minus 0.50), +0.256 [+0.184, +0.339] (0.50 minus 0.75), and +0.326 [+0.233, +0.422] (0.75 minus random-valid).

Three deviations are documented. Generation reuses the frozen candidate generator, which reproduces all 12 snapshots byte-for-byte although domain generators take the seed differently. A first concurrent generation run corrupted the blocksworld outputs through a shared temporary file, so it was discarded, all candidates were regenerated serially and reproduced, and the deviation was disclosed before evaluation. Only the two additive native references seed the pre-rendered views, rather than four algorithms, because native BFS exceeds the 128-expansion reference ceiling on these tasks, and unseen states are rendered live.

Table 16: The frozen validation panel (12 tasks, 7 domains). R is the uncapped exact-reference expansion count (greedy / w3), and Screen counts random-valid goals over 10 screening episodes.

Task	C^*	R	Screen
15puzzle-935006	8	46 / 12	1
blocksworld-935001	10	26 / 26	4
blocksworld-935004	10	13 / 13	2
depot-935005	10	14 / 15	1
depot-935011	9	13 / 13	1
elevators-935000	8	11 / 11	2
elevators-935013	8	11 / 11	2
grid-935016	8	9 / 9	4
storage-935000	10	20 / 17	1
storage-935001	8	18 / 15	2
towers_of_hanoi-935000	13	24 / 25	1
towers_of_hanoi-935002	15	24 / 26	2

Table 17: All arms on the validation panel. M1 carries a task-clustered 95% interval (12-task bootstrap, seed 133, 10,000 draws), and M2 and M3 are solved-only. Only the five ladder arms enter the pre-registered test. The descriptive selectors read privileged information, and the base emits no valid first choice under its one-call cap.

Arm	M1 [95% CI]	$2\times$	M2 ρ	M3 κ
Exact reference	0.875 [0.875, 0.875]	1	1.0	1.021
Exact- ϵ 0.25	0.793 [0.764, 0.824]	0.992	1.089	1.024
Exact- ϵ 0.50	0.603 [0.528, 0.670]	0.95	1.246	1.032
Exact- ϵ 0.75	0.347 [0.252, 0.445]	0.717	1.374	1.023
Random-valid	0.021 [0.007, 0.038]	0.083	1.692	1.042
hadd-greedy	0.882 [0.774, 0.955]	0.958	0.962	1.083
bfs-order	0.000 [0.000, 0.000]	0	undefined	undefined
Novelty-first selection	0.021 [0.000, 0.052]	0.167	1.909	1.000
worst-first	0.000 [0.000, 0.000]	0	undefined	undefined
First adapter	0.026 [0.010, 0.042]	0.208	1.810	1.050
Scaled adapter (seed 17)	0.349 [0.172, 0.531]	0.583	1.326	1.140
Pretrained base	0.000 [0.000, 0.000]	0	undefined	undefined

U RESULTS DETAIL: SCALED ADAPTER AND SEED REPLICATION

The protocol was frozen before task generation, the corpus before training, and the schedule before launch. D is the mean over training seeds of learned M1 minus random-valid M1, with a task-cluster interval (10,000 draws, seed 133). POSITIVE requires a positive lower bound, $D \geq 0.05$, and a positive estimate at every seed, and the equivalence margin is ± 0.05 . Controls are reused from the ladder test.

Of 2,400 serially generated training tasks, 1,182 remained eligible after excluding overlap with every evaluation panel and deduplicating, and the seed-major walk reached 2048 records per algorithm from 236 tasks in 12 domains. Two reseeded menu orders per record give 8,192 samples, with the target last in 17.58% of samples against a chance rate of 17.58%.

The recipe matches the first adapter’s except for corpus, augmentation, sample order, and seed (LoRA rank 64, alpha 128, 128 steps, seed 17). The smoke gate accepted 34/34 calls, and all 24/24 learned episodes, capped at $2R$, replayed with no mismatch and no invalid operation. For seed 17, the $2\times$ solve fraction is 0.583, M2 ρ 1.326, and M3 κ 1.140. Over 691 multi-option decisions heap-head agreement is 0.210 against chance 0.125, the last-label rate is 0.205, and the first-label rate is 0.090.

Four choices are documented. The scope edit to one training seed left seed variance unmeasured in that run. A later pre-registered replication trained seeds 29 and 71 with the byte-identical recipe and the sample order pinned to the seed-17 order, so seeds change only the LoRA initialisation and dropout, and all 48 of its new episodes were independently replayed with no mismatch. The step budget matches the first adapter’s recipe. The seed-major corpus walk resolves an ambiguity in the protocol text. Both evaluation hooks first failed because the scheduler passes only the terminal path, and after a fix deriving the attempt directory from it, the audit re-ran on the same attempts and passed without affecting GPU work or episodes.

On the validation panel, D is $+0.328$ [$+0.155, +0.508$] for seed 17, $+0.229$ [$+0.108, +0.354$] for seed 29, and $+0.297$ [$+0.134, +0.466$] for seed 71. Per-seed M1 is 0.349, 0.250, and 0.318, with a between-seed standard deviation of 0.051 (descriptive). The last-label rate per seed is 0.205, 0.192, and 0.171 (descriptive). The pre-registered separation from the exact- ϵ 0.75 rung is $S = +0.002$,

Table 18: Per-task D (scaled adapter M1 minus random-valid M1) for seed 17 on the validation panel. The pre-registered three-seed endpoint is $D_3 = +0.285 [+0.140, +0.433]$ (Table 20).

Task	D	Task	D
15puzzle-935006	-0.0625	elevators-935013	0.6875
blocksworld-935001	0.0	grid-935016	0.3625
blocksworld-935004	0.0	storage-935000	0.7375
depot-935005	0.0625	storage-935001	0.7
depot-935011	0.7125	towers_of_hanoi-935000	0.2375
elevators-935000	0.5	towers_of_hanoi-935002	0.0

Table 19: M1 with task-clustered 95% intervals on the validation panel (12 tasks), the held-out panel P2 (11 tasks), and the unscreened panel P2u (12 tasks). All entries are descriptive. Ladder rungs and hadd-greedy read privileged information and are not realisable policies.

Arm	Validation (12)	P2 (11)	P2u (12)
Exact reference	0.875 [0.875, 0.875]	0.875 [0.875, 0.875]	0.875 [0.875, 0.875]
Exact- ϵ 0.25	0.793 [0.764, 0.824]	0.745 [0.685, 0.805]	0.678 [0.632, 0.721]
Exact- ϵ 0.50	0.603 [0.528, 0.670]	0.507 [0.445, 0.566]	0.397 [0.331, 0.478]
Exact- ϵ 0.75	0.347 [0.252, 0.445]	0.205 [0.145, 0.269]	0.149 [0.091, 0.227]
Random-valid	0.021 [0.007, 0.038]	0.016 [0.000, 0.048]	0.002 [0.000, 0.006]
Pretrained base	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
Adapter, seed 17	0.349 [0.172, 0.531]	0.506 [0.313, 0.699]	0.198 [0.031, 0.396]
Adapter, seed 29	0.250 [0.125, 0.375]	0.284 [0.119, 0.489]	0.120 [0.026, 0.224]
Adapter, seed 71	0.318 [0.151, 0.490]	0.506 [0.295, 0.705]	0.135 [0.031, 0.250]
Adapter, three-seed mean	0.306 [0.158, 0.455]	0.432 [0.273, 0.602]	0.151 [0.030, 0.288]
hadd-greedy (privileged)	0.882 [0.774, 0.955]	0.928 [0.883, 0.966]	0.906 [0.883, 0.933]

-0.097, and -0.029 per seed. Against the 0.50 rung the three-seed S is -0.298 [-0.468, -0.129] (descriptive). Table 20 sets these values beside the held-out and unscreened panels.

U.1 HELD-OUT PANELS

Both fresh panels come from 2,400 candidates generated from seeds 955000–955199. Overlap with earlier panels removed 1,168 candidates, duplicates 205, goal at the initial state 31, and C^* out of range 536, which left 460. P2 applies the validation panel’s generation, C^* , and screening rules. A candidate is kept when $8 \leq C^* \leq 20$ and random-valid reaches the goal in 1 to 9 of its 10 screening episodes, and the freeze walk, sorted by domain and seed, takes at most 2 tasks per domain up to 12 tasks, with a minimum of 8 tasks from 5 domains. P2 froze with 11 tasks from 8 domains because only 11 admitted tasks were available under the per-domain limit. It met the minimum, so every P2 test is confirmatory. P2u takes kept candidates outside P2 on which the exact reference reaches the goal under both algorithms, without consulting the random-valid screen, at most 2 per domain, giving 12 tasks from 7 domains. All 12 P2u tasks happen to have 0/10 random-valid goals in screening. All 638/638 P2 and 696/696 P2u zoo episodes were independently replayed, with none missing.

The first freeze, from the original 40-seed pool 955000–955039, admitted 6 tasks from 5 domains and took the protocol’s blocked exit. Only controls-only screening had run, and no evaluation episode existed. An amendment then extended the seeds to 955000–955199. It kept the 480 earlier candidates unchanged and generated the new seeds 955040–955199 serially after them. The generation mechanism, the exclusion sources, the duplicate check, the C^* rule, R_t , the screen, the freeze walk, and the P2u rule stayed unchanged, and both panels were frozen from the combined pool before any evaluation episode.

Table 19 gives each arm’s M1 on the three panels. The four pre-registered adjacent gaps on P2 are +0.130 [+0.070, +0.190], +0.239 [+0.183, +0.290], +0.302 [+0.259, +0.340], and +0.189 [+0.136, +0.248], so the ladder test passes. On P2u the descriptive gaps are +0.197 [+0.154, +0.243], +0.281 [+0.191, +0.355], +0.248 [+0.209, +0.290], and +0.147 [+0.088, +0.226], and every lower bound is positive.

Table 20 gives D per seed and D_3 with intervals on each panel and pooled over 35 tasks, and the separation S from the exact- ϵ rungs. Seed 29 has the lowest per-seed D on every panel, and on P2 its separation point from the 0.75 rung is +0.080 (descriptive). No equivalence margin was pre-registered for S , so a non-separated S is not equivalence to the rung. The P2 separation rests on 11 task clusters, with 8 positive and 3 negative per-task differences, and percentile intervals can under-cover at this size.

Table 20: Per-seed D and three-seed D_3 (adapter M1 minus random-valid M1), and three-seed separation S (adapter M1 minus exact- ϵ rung M1), with paired task-clustered 95% intervals (10,000 draws, seed 133). Pre-registered entries are marked pre-reg. and the rest are descriptive.

	Validation (12)	P2 (11)	P2u (12)	Pooled (35)
D , seed 17	+0.328 [+0.155, +0.508]	+0.490 [+0.294, +0.688]	+0.196 [+0.031, +0.396]	+0.334 [+0.223, +0.450]
D , seed 29	+0.229 [+0.108, +0.354]	+0.268 [+0.093, +0.477]	+0.118 [+0.026, +0.220]	+0.203 [+0.124, +0.290]
D , seed 71	+0.297 [+0.134, +0.466]	+0.490 [+0.276, +0.694]	+0.133 [+0.031, +0.248]	+0.301 [+0.200, +0.411]
D_3	+0.285 [+0.140, +0.433]	+0.416 [+0.250, +0.591]	+0.149 [+0.030, +0.283]	+0.279 [+0.191, +0.375]
D_3 verdict	POSITIVE (pre-reg.)	POSITIVE (pre-reg.)	POSITIVE (pre-reg.)	POSITIVE (descriptive)
S vs 0.75	-0.041 [-0.187, +0.102]	+0.227 [+0.053, +0.403]	+0.002 [-0.145, +0.157]	+0.058 [-0.040, +0.156]
S verdict	NOT_SEPARATED (pre-reg.)	SEPARATED_ABOVE (pre-reg.)	NOT_SEPARATED (descriptive)	NOT_SEPARATED (descriptive)
S vs 0.50	-0.298 [-0.468, -0.129]	-0.075 [-0.247, +0.089]	-0.246 [-0.400, -0.082]	-
S verdict	SEPARATED_BELOW (descriptive)	NOT_SEPARATED (descriptive)	SEPARATED_BELOW (descriptive)	-

Table 21: Concentration of the three-seed difference (descriptive). Top, tasks by the sign of the per-task difference. Bottom, D_3 over the validation panel and P2 with one domain left out, and the number of tasks left.

Panel	Positive	Zero	Negative
Validation (12)	9	2	1
P2 (11)	9	2	0
P2u (12)	4	8	0
Union (35)	22	12	1
Domain left out	D_3	95% interval	Tasks left
15-puzzle	+0.384	[+0.269, +0.501]	21
blocksworld	+0.395	[+0.269, +0.520]	19
depot	+0.353	[+0.224, +0.484]	20
elevators	+0.305	[+0.185, +0.430]	20
ferry	+0.331	[+0.212, +0.455]	21
grid	+0.331	[+0.204, +0.464]	20
storage	+0.332	[+0.209, +0.459]	21
Tower of Hanoi	+0.384	[+0.262, +0.507]	20
visitall	+0.318	[+0.211, +0.422]	22

Table 21 counts tasks by the sign of their three-seed difference and leaves out one domain at a time over the validation and held-out panels. Over the 35-task union 22 tasks are positive, 12 are zero, and 1 is negative. On P2u the eight zero tasks are both 15-puzzle, both blocksworld, both depot, and both driverlog tasks, on which neither the adapter nor random-valid solves within $2R_t$. Every leave-one-domain-out D_3 lies from +0.305 to +0.395, and every lower bound is at least +0.185. All of these results are descriptive.

U.2 DAGGER ABLATION

The pre-registered ablation collects on-policy records with the seed-17 adapter under each additive algorithm, aggregates them with replay records from the frozen corpus, retrain, and evaluates on the validation panel and the held-out panel P2. The primary endpoint is the paired separation S of the DAGger adapter from the exact- ϵ 0.75 rung pooled over the 23 tasks, with SEPARATED_ABOVE requiring a positive lower bound. The co-primary is the change Δ in M1 from the pre-DAGger adapter, with equivalence margin ± 0.05 .

An amendment at 11:21 UTC on 2026-09-24, decided before any DAGger training or evaluation outcome, cut the plan to seed 17 because the three-seed plan could not fit a hard wall-clock deadline. The running seed-29 collection was cancelled with 1.79 GPU-h still charged, and seed 71 was not run. Each algorithm’s training set became the first 512 on-policy seed-17 records plus 512 replay records, which the menu augmentation turns into 2048 samples over 64 steps, with the recipe otherwise unchanged.

All 24/24 validation-panel and 22/22 P2 DAGger episodes were independently replayed with no mismatch. The primary endpoint is NOT_SEPARATED, $S = +0.115 [-0.026, +0.258]$ over 23 tasks, and the co-primary is INCONCLUSIVE, $\Delta = -0.030 [-0.114, +0.057]$. As descriptive secondary results, S is +0.278 [+0.082, +0.473] on P2 and -0.034 [-0.197, +0.135] on the validation panel, and the last-label rate on the validation panel falls from 0.205 to 0.137. One small round at one seed therefore produces no detectable change in measured choice quality, since Δ is inconclusive against the ± 0.05 margin, and the ablation is inconclusive.

Table 22: Zero-shot base against random-valid (D_3), the exact- ϵ 0.75 rung (S), and the three-seed adapter (A), with paired task-clustered 95% intervals (10,000 draws, seed 133) and exact sign-flip p values (descriptive). Panel endpoints are pre-registered and pooled entries are descriptive.

	Validation (12)	P2 (11)	Pooled (23, descriptive)
D_3	+0.005 [−0.025, +0.044]	+0.001 [−0.048, +0.051]	+0.003 [−0.024, +0.033]
verdict, p	EQUIVALENT, 0.906	INCONCLUSIVE, 1.000	EQUIVALENT, 0.896
S vs 0.75	−0.321 [−0.422, −0.226]	−0.188 [−0.239, −0.140]	−0.257 [−0.324, −0.197]
verdict, p	SEPARATED_BELOW, 4.88×10^{-4}	SEPARATED_BELOW, 9.77×10^{-4}	SEPARATED_BELOW, 2.38×10^{-7}
A vs adapter	−0.280 [−0.425, −0.142]	−0.415 [−0.589, −0.248]	−0.344 [−0.459, −0.232]
verdict, p	SEPARATED_BELOW, 0.004	SEPARATED_BELOW, 0.004	SEPARATED_BELOW, 7.63×10^{-6}

Table 23: The seven surveyed published interfaces, with the survey’s pre-audit prediction for random-valid against the reference.

Interface	Venue	Type	CPU-feasible	Prediction
AutoPlanBench (Stein et al., 2025)	ICAPS 2025	B	yes	divergent
ScienceWorld (Wang et al., 2022)	EMNLP 2022	B	yes	divergent
GLAM (BabyAI-Text)	ICML 2023	B	yes	divergent
WebShop	NeurIPS 2022	B	no	divergent
RAP	EMNLP 2023	C	no	divergent
LLM-First Search (Herr et al., 2025)	arXiv 2025	C	no [†]	saturated [†]
EmbodiedBench	ICML 2025	B	no	divergent

[†] Audited later with a GPU reference model, and the prediction held under the paper’s budget (Appendix V.1).

U.3 ZERO-SHOT POLICY

The second realisable policy is the pretrained Qwen3-VL-8B-Instruct base with no adapter, given the frozen scene-only choice-frontier observation, one appended system sentence asking for a single JSON choice, a fixed three-rule output extractor, greedy decoding at inference seed 17, and the adapter’s cap of $2R_t$ calls. It ran on both additive algorithms over the validation panel and P2, 46 episodes in all. The preferred arm, InternVL3.5-8B trained with the frozen three-seed recipe, was ruled out before any outcome. Its mean observation takes 11,748 tokens against 3,111 for Qwen3-VL, which puts training alone at 43 to 55 GPU-h against the 30 GPU-h cap, and a reduced recipe would break the byte-identical condition, so it was not run in any form. The protocol was committed before any GPU launch, and no amendment followed.

Table 22 gives the endpoints. On P2 the upper bound of D_3 exceeds the ± 0.05 margin by 0.001, so the frozen rule returns INCONCLUSIVE rather than EQUIVALENT. The arm’s M1 is 0.026 [0.000, 0.063] on the validation panel, 0.017 [0.000, 0.051] on P2, and 0.022 [0.003, 0.046] pooled, below the exact- ϵ 0.75 rung on every panel. All 1,592 model calls produced a valid choice through the first extractor rule. Heap-head agreement minus chance is +0.003 and +0.001, the last-label rate is 0.328 and 0.285 against a chance rate of about 0.13 to 0.14, and 4 of 46 episodes solve at $2R_t$ (descriptive). The pre-registered prediction, D_3 within the margin with S and the adapter contrast A separated below on both panels, held. All 46/46 episodes were independently replayed with no mismatch.

V RESULTS DETAIL: EXTERNAL IDENTITY AUDIT

The survey admitted published evaluations from 2022 on in which the model chooses from a runtime-supplied candidate set inside a planning or search loop, with public tasks and code. Seven candidates passed, five of type B (the model picks one valid action per step), two of type C (the model scores candidates that a search orders), and none of type A, our complete-set submission contract (Table 23). Four are not CPU-feasible under their papers’ budgets. For WebShop (free-text search, no step limit) and LLM-First Search (an unreported token cap) this was a judgement call, and LLM-First Search was the only saturated prediction, which the later audit in the LLM-First Search Audit appendix (Appendix V.1) tested. ScienceWorld is the most cited feasible candidate, and the runner-up shares its prediction.

The frozen protocol takes the first 200 test variations under the paper’s 100-step limit with five random-valid seeds (17, 5077, 6131, 7409, and 8527), and ten re-run episodes are byte-identical. The reference completes 183 of 200 (0.915), including 3/9 boil, 6/9 melt, 4/9 freeze, 6/9 change-state, 135/135 use-thermometer, and 29/29 measure-melting-point variations, and falls short only where 17 of the 36 Matter variations need more than 100 moves. Random-valid completes none, and 997 of its 1000 episodes end in a failure score.

Table 24: LLM-First Search success by configuration under the paper’s token budget and at the reference’s own number of expansions. Differences are reference minus random-valid with task-cluster 95% intervals (10,000 draws, seed 133, descriptive). The reference is the substituted local model.

Configuration	Reference	Paper token budget		Matched decision budget	
		Random-valid	Difference	Random-valid	Difference
Countdown 3 (20)	1.000	1.000	0.000 [0.000, 0.000]	0.120	+0.880 [+0.770, +0.970]
Countdown 5 (20)	0.750	1.000	−0.250 [−0.450, −0.100]	0.110	+0.640 [+0.370, +0.870]
Countdown 7 (20)	0.650	1.000	−0.350 [−0.550, −0.150]	0.020	+0.630 [+0.410, +0.840]
Sudoku 4×4 (20)	0.900	1.000	−0.100 [−0.250, 0.000]	0.060	+0.840 [+0.700, +0.950]
All (80)	0.825	1.000	−0.175 [−0.263, −0.100]	0.0775	+0.747 [+0.648, +0.838]
Verdict			SATURATED		CHOICE_REGISTERED

The slice covers 6 of 30 task types (200 of 1,819 test variations), on all of which the paper’s own Random-Valid score is 0.00. The paper reports normalized scores, so its DRRN score of 0.084 on these tasks is not a success rate. The audit uses ScienceWorld 1.3.0, whose notes warn that gold paths can change relative to 1.2.x, and the paper’s simulator version was not re-run.

V.1 LLM-FIRST SEARCH AUDIT

LLM-First Search is a type-C interface. The model scores candidates, and the search expands the argmax child or pops the highest-valued frontier node, so every unexpanded alternative stays in the frontier. Random-valid answers every query with independent uniform draws, spends zero tokens, and makes no model call, and a decision is one node expansion after the root. The upstream code (commit 3025bdaa, Apache-2.0) is imported unchanged and not vendored. The fixed rule audits every configuration whose published GPT-4o win rate exceeds 10%, which excludes Sudoku 6×6 and keeps Countdown 3, 5, and 7 and Sudoku 4×4, with games 0 to 19 of each, 80 tasks in all. Budgets are the paper’s per-game token caps (1,000,000 for Countdown and 100,000 for Sudoku 4×4). Four deviations are declared. The paper’s GPT-4o is replaced by the local Qwen3-30B-A3B-Instruct-2507, so reference success is that model’s, the reference runs once per task, Sudoku 6×6 is excluded, and random-valid uses a faster frontier implementation that produced byte-identical logs on 57/57 probe episodes. Random-valid uses the five seeds of the first audit.

All 80/80 reference and 400/400 random-valid episodes were independently replayed with no mismatch. No pair of the 400 is identical, and the first divergence has median index 0 and maximum 2. Table 24 gives both readings. Under the paper’s budget random-valid wins every episode at every seed, so the verdict is SATURATED, and every reference loss ends at the token cap. Within the reference’s own number of expansions per task, random-valid wins 0.0775 (per seed 0.0625, 0.075, 0.075, 0.0875, and 0.0875), so the verdict is CHOICE_REGISTERED. Both pre-registered predictions held. Published GPT-4o success on these configurations is 0.768. One amendment, made before any audit episode, only reordered the smoke and audit start, and the ledger spent 2.89 of 12 GPU-h with no paid API use.

W RESULTS DETAIL: SEED REPLICATION AND FOLLOW-UP LIMITS

The prospective training-seed replication (seeds 17/29/71 frozen pre-launch, 534 new episodes independently replayed) executed on two cells (Table 4). In the greedy × multimodal replication cell (greedy best-first × multimodal-state, 24-problem unseen panel), per-seed success is 21/24, 21/24, 22/24 (mean 0.889, range [0.875, 0.917]). The contrast against the pretrained base is positive at every seed (+0.875/+0.875/+0.917, no sign flip). The contrast against random-valid stays negative (−0.125/−0.125/−0.083). On the DAgger iteration-1 cell set (BFS × three modalities, DAgger and continued-SFT arms), every arm and seed scores 0/72 unseen, with paired contrast +0.000 [+0.000, +0.000] at each seed. The published iteration-2 verdict (1/72 vs 1/72, seed 17) is unchanged. Both cells replicate across all three seeds. Cell A covers one of the three modalities of the greedy additive cell. The remaining two modalities and all other enumeration-contract cells use seed 17 only. Cross-seed aggregation is descriptive ($n=3$). The image-only-arm proposal has since executed under the native-arms v2 amendment (the Image-Only Native Arms appendix, Appendix Q), and its v1 gate remains as history.

Table 25: The two decision contracts. Rows compare what the policy emits, who owns the frontier, what the observation carries, how the budget binds, and the role of the identity audit. Results under the two contracts are not comparable as performance rows.

	Enumeration contract	Choice-frontier contract
Policy emits	One Typed Search Operation per step, taken from a grounded, unscored candidate action menu	One opaque label from a menu of frontier states, naming the state to expand next
Frontier	Kept in textual Search Memory with visited and score state, advanced by submitted operations	Owned by the runtime. Menu labels are re-assigned per decision by a seeded permutation
Observation	Set by the Modality Observation adapter	Visual only, with task pages, the initial scene, one unlabelled 128px scene per frontier state, and goal pages
Budget	Learned arms capped at $2\times$ the exact-reference decision count, base at one call	Decisions and expansions capped at $2\times$ the additive exact-reference expansion count, goal selection uncharged
Identity audit	Diagnostic run on the expanded-baseline additive cells	Pre-registered pass condition over 18 control pairs

The follow-up windows share this development-stage status, and we list their limits separately. The native-arms window evaluates an 18-unit clean panel with one training seed per cell. The R1-R4 follow-up window reuses the frozen native-arms adapters for R1, R2, and R4, and trains one seed-17 run per cell for R3. The choice-frontier window trains one seed per algorithm on a 9-task panel whose discriminative subset holds one task. The comparator zoo is a CPU-only exploratory analysis. The scaled adapter has three training seeds on the validation panel, the held-out panel P2, and the unscreened panel P2u, and the DAgger ablation trains one seed on half the planned corpus. None of these windows accessed the frozen 45-task held-out manifest. The shared nine-task panel draws from 8 domains with 7 compact and 2 expanded variants.

Final primary evaluation prerequisites. No authorized held-out final evaluation on the frozen 45-task manifest exists. Freezing one requires recorded decisions on the following prerequisites. (1) The condition-level data source and Search-Trace Segment construction with per-algorithm allocation. (2) Compute parity and checkpoint-selection rules. (3) The episode-budget unit, the whole-instance split procedure, and the fixed Search Memory capacity. (4) The estimand, the uncertainty interval or test, the success threshold, the invariant-compliance rule, and the validity and invalid-operation denominators for every cell. Until these choices are recorded, the held-out final evaluation cannot be frozen and no final comparative efficacy finding exists.

X DECISION CONTRACTS

Table 25 compares the enumeration and choice-frontier contracts defined in Section 4.

Y USE OF LLM STATEMENT

An LLM was used only for polishing language; the scientific content is entirely original.